

Unit 3

CHEMICAL BONDING

107 items · 320 marks

Syllabus sections in this unit

- 3.1 Electronegativity and bonding
- 3.2 Ionic bonding
- 3.3 Metallic bonding
- 3.4 Covalent bonding and coordinate (dative covalent) bonding
- 3.5 Shapes of molecules
- 3.6 Intermolecular forces, electronegativity and bond properties
- 3.7 Dot-and-cross diagrams

Unit 3: Chemical bonding

1. M/J 26/P21/Q1/a,c,d

- 1 (a) (i) Complete the dot-and-cross diagram in Figure 1.1 to show the arrangement of outer shell electrons in a molecule of SF_6 .

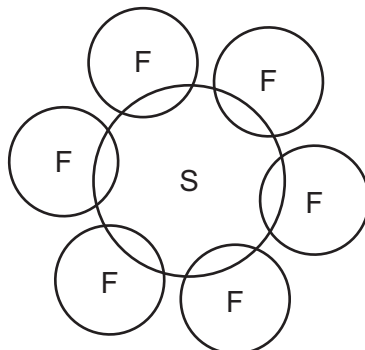


Figure 1.1

[1]

- (ii) Name the shape of a molecule of SF_6 .

[1]

- (c) A molecule of SCl_2 is polar.

- (i) Predict bond angle x in Figure 1.2 using VSEPR theory. Explain your answer.

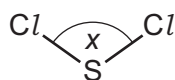


Figure 1.2

bond angle $x =$

explanation

[3]

- (ii) Show on Figure 1.3 all the dipoles present in the molecule of SCl_2 .

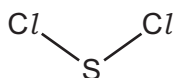


Figure 1.3

[1]

- (d) SCl_2 and H_2O both contain instantaneous dipole-induced dipole attractions and permanent dipole attractions.

- (i) Complete Table 1.2.

Table 1.2

	SCl_2	H_2O
total number of electrons per molecule		10

[1]

- (ii) Identify which molecule, SCl_2 or H_2O , has the stronger instantaneous dipole–induced dipole attractions. Explain your answer.

[1]

- (iii) H_2O has a greater dipole moment than SCl_2 . Explain why.

[1]

2. M/J 26/P22/Q1/b

- 1** Oxygen reacts with many elements.

- (b) Lithium reacts with oxygen to produce lithium oxide, Li_2O .

Li_2O is a white solid with a melting point over 1400°C .

Draw a dot-and-cross diagram to show the arrangement of **all** the electrons in Li_2O .

[2]

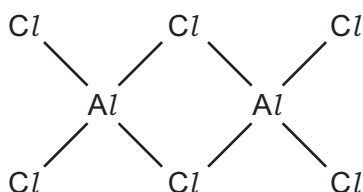
3. M/J 26/P24/Q1/c,d

- 1** Aluminium is a Group 13 element that forms a number of compounds.

Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

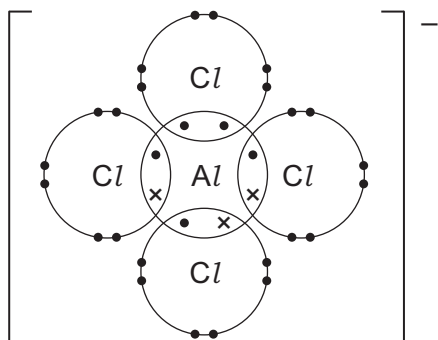
- (c)** AlCl_3 dimerises to form Al_2Cl_6 . Al_2Cl_6 is shown in Figure 1.2.

**Figure 1.2**

Describe how coordinate (dative covalent) bonds form when two AlCl_3 molecules combine to form Al_2Cl_6 .

[1]

- (d)** Aluminium forms the AlCl_4^- ion. The dot-and-cross diagram for AlCl_4^- in Figure 1.3 shows the arrangement of outer electrons.

**Figure 1.3**

- (i)** Use Figure 1.3 and VSEPR theory to name the shape of the AlCl_4^- ion and to predict the $\text{Cl}-\text{Al}-\text{Cl}$ bond angle.

name of shape

bond angle

[2]

- (ii)** Draw the three-dimensional shape of the AlCl_4^- ion.

[1]

4. M/J 26/P24/Q2/a(i)

- 2 (a) (i) Nitrogen, N_2 , contains sigma (σ) and pi (π) bonds.

Describe how a pi (π) bond is formed.

[2]

5. MAR 26/P22/Q2/a(ii)

- 2 Nitrogen monoxide, NO, is an atmospheric pollutant.

- (a) Figure 2.1 shows the dot-and-cross diagram of NO.

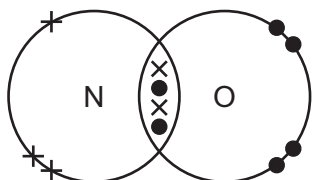


Figure 2.1

- (ii) In NO, the N and O atoms are bonded by a sigma (σ) and a pi (π) bond.

Complete Figure 2.2 to show the pi bond in NO that is formed from orbital overlap.



Figure 2.2

[1]

6. MAR 26/P22/Q4/b

- 4 Cyanoethanoic acid, $CH_2(CN)COOH$, is a weak organic acid.

- (b) $CH_2(CN)COOH$ is very soluble in water because it can form hydrogen bonds with water.

Complete Figure 4.2 to show **one** hydrogen bond between a molecule of $CH_2(CN)COOH$ and a molecule of H_2O .

Label the hydrogen bond clearly, and show any relevant partial charges and lone pairs of electrons.

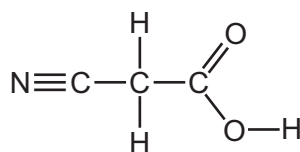


Figure 4.2

[3]

10. M/J 25/P23/Q4/b

- 4 The reaction between $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{HCl}(\text{aq})$ is monitored at constant temperature.

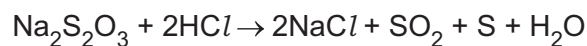


Fig. 4.1 shows how the concentration of $\text{HCl}(\text{aq})$ varies with time.

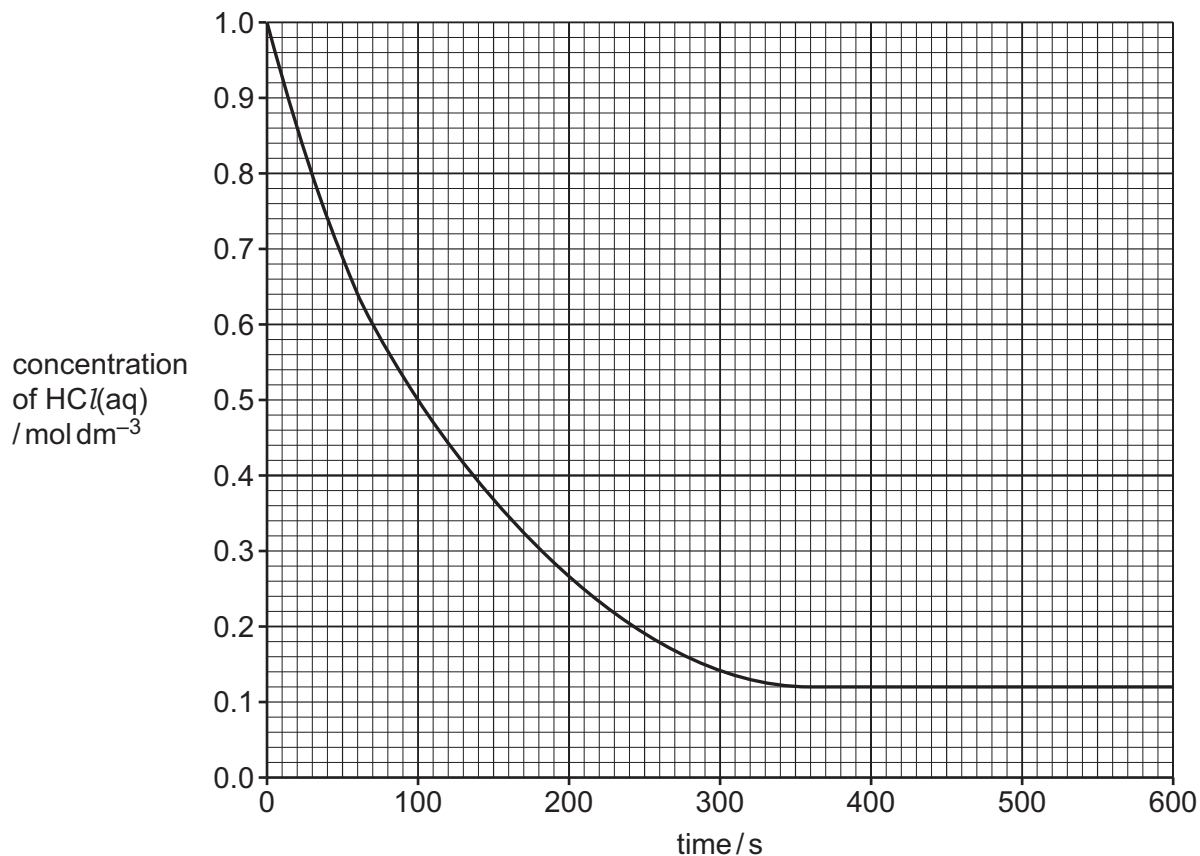


Fig. 4.1

- (b) Fig. 4.2 shows a possible arrangement of outer-shell electrons in one SO_2 molecule.

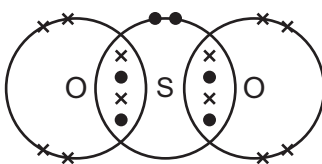


Fig. 4.2

- (i) Use Fig. 4.2 to predict the shape and bond angle of a molecule of SO_2 .

shape

bond angle

°

[2]

- (ii) Use Table 4.1 to predict the strength of the dipole moment of SO_2 , if any, compared to that of H_2O . Explain your answer.

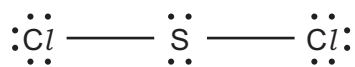
Table 4.1

	H	O	S
electronegativity	2.1	3.5	2.6

[2]

11. M/J 25/P24/Q2/a

- 2 (a) Fig. 2.1 shows the covalent bonds and lone pairs of electrons in a molecule of SCl_2 .

**Fig. 2.1**

- (i) Use VSEPR theory to predict the shape of a molecule of SCl_2 .

[1]

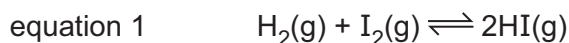
- (ii) Predict the bond angle in a molecule of SCl_2 .

°

[1]

12. M/J 25/P24/Q3/d(i)

- 3 $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.



In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

- (d) The value of K_c for the dissociation of HCl at 298 K is 5.50×10^{-34} .



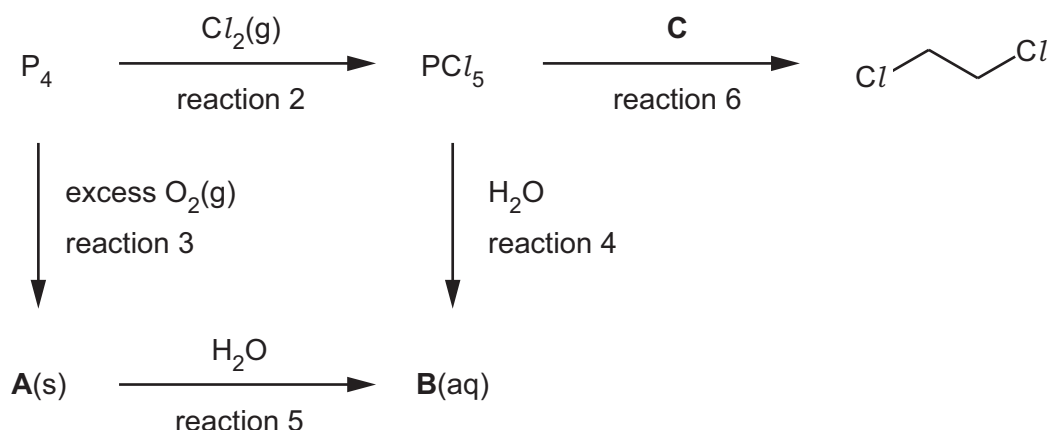
- (i) Define covalent bond.

[1]

13. MAR 25/P22/Q1/b(ii)

- 1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

- (b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2 .

**Fig. 1.1**

- (ii) Predict the shape of a molecule of PCl_5 .

[1]

14. MAR 25/P22/Q2/a

- 2 (a) The Group 2 elements show metallic bonding.

Define metallic bonding.

[2]

15. O/N 24/P21/Q3/a(iii),d

- 3 The Group 14 elements show a change from non-metallic to metallic character down the group.

- (a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

- (iii) Explain why Sn has good electrical conductivity.

[1]

- (d) Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

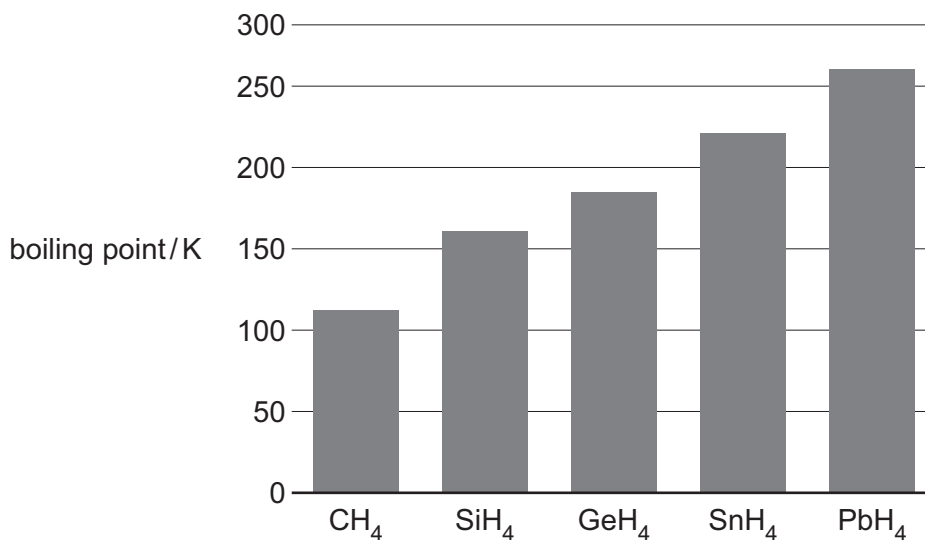


Fig. 3.2

- (i) Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

[2]

- (ii) Deduce the shape of a molecule of SiH₄.

[1]

16. O/N 24/P22/Q2/a(ii),c

2 Oxygen is a Group 16 element.

(ii) Draw a dot-and-cross diagram to show the species present in Al_2O_3 .

Draw outer electrons only.

[1]

(c) Fig. 2.1 shows the boiling points of H_2O and other Group 16 hydrides.

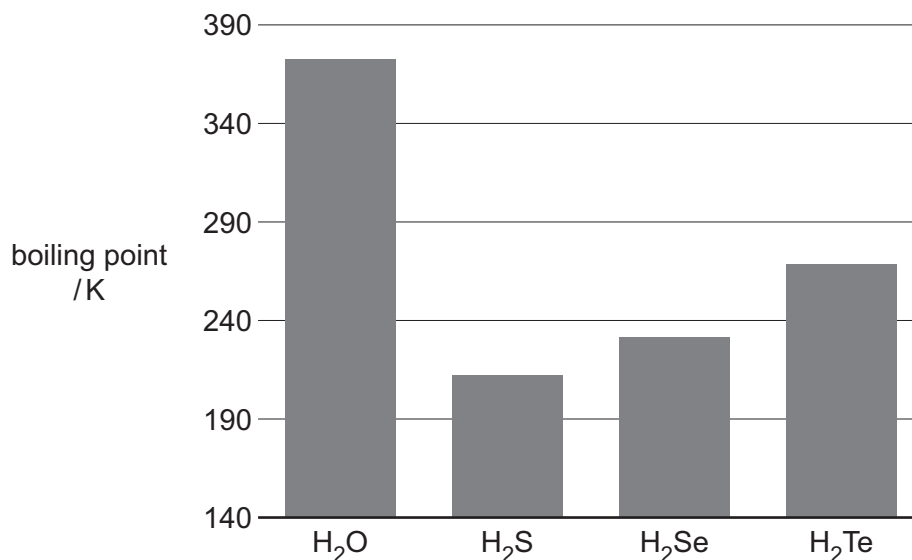


Fig. 2.1

(i) Explain the trend in the boiling points of the Group 16 hydrides H_2S to H_2Te .

[2]

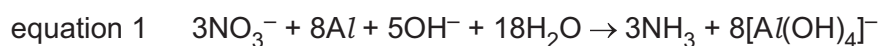
(ii) Explain why the boiling point of H_2O is much higher than that of H_2S .

[1]

17. O/N 24/P22/Q3/b(iv),f(i)

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(b) A common test for nitrates is the reaction with NaOH and Al . Equation 1 shows the reaction.



(iv) Suggest the shape of the $[\text{Al}(\text{OH})_4]^-$ ion.

[1]

(f) At very high temperatures, phosphorus can form P_2 molecules.

P_2 contains a triple bond, $\text{P}\equiv\text{P}$.

(i) Describe the formation of the $\text{P}\equiv\text{P}$ bond in terms of orbital overlap.

[2]

18. O/N 24/P23/Q3/a(iii),d

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

- (a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

- (iii) Explain why Sn has good electrical conductivity.

[1]

- (d) Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

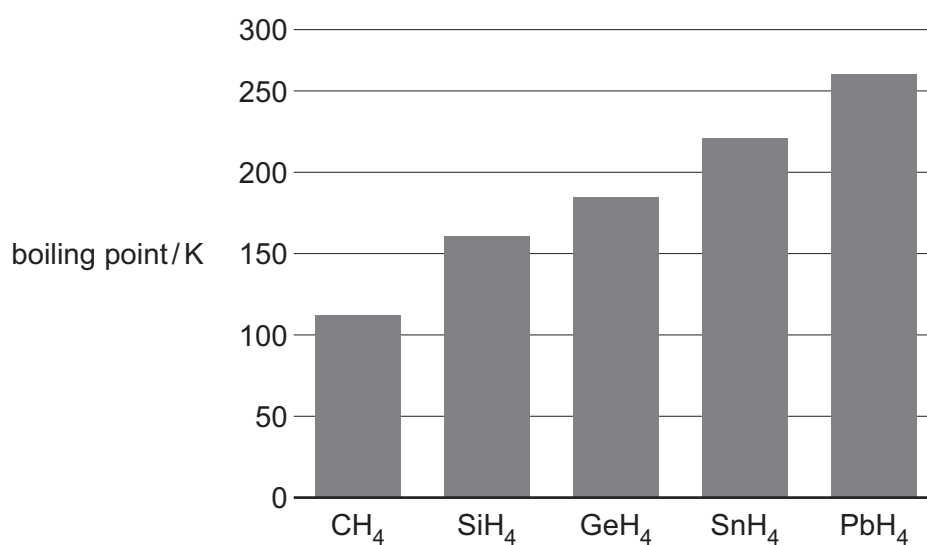


Fig. 3.2

- (i) Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

[2]

- (ii) Deduce the shape of a molecule of SiH₄.

[1]

19. M/J 24/P21/Q2/b

(b) A molecule of SCl_2 contains two S–Cl covalent bonds.

- (i) Complete the dot-and-cross diagram in Fig. 2.1 to show the arrangement of the outer electrons in a molecule of SCl_2 .

Use $\times$ to show electrons from the chlorine atoms.

Use $\bullet$ to show electrons from the sulfur atom.

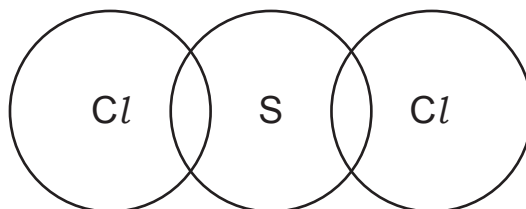


Fig. 2.1

[2]

- (ii) Predict the shape of, and bond angle in, a molecule of SCl_2 by using VSEPR theory.

shape

bond angle

[2]

20. M/J 24/P21/Q5/a

5 Hydrocarbon molecules contain covalent bonds.

- (a) Define covalent bond.

[1]

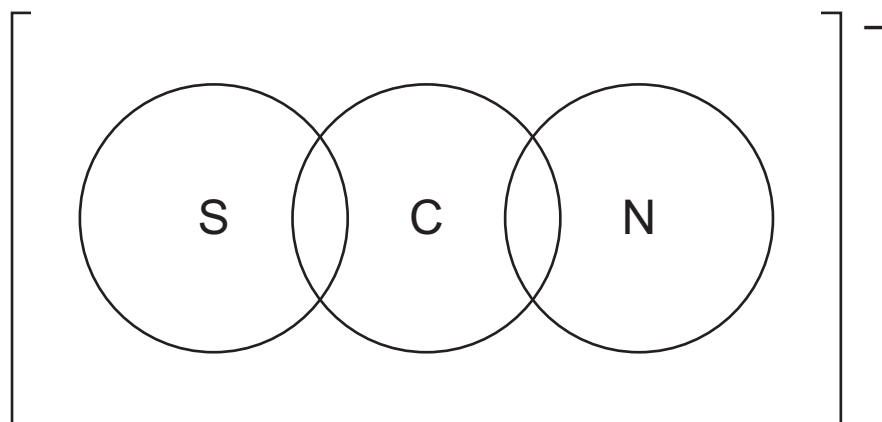
21. M/J 24/P22/Q1/c

- (c) Draw a labelled diagram to show the structure and bonding in sodium.

[1]

22. M/J 24/P22/Q3/d**(d)** $\text{SCN}^-(\text{aq})$ is colourless.

Complete the dot-and-cross diagram in Fig. 3.1 to show the arrangement of outer electrons in an SCN^- ion.

**Fig. 3.1**

[2]

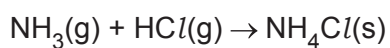
23. M/J 24/P23/Q1/a(ii)**(ii)** Covalent bonds can be σ bonds or π bonds.

Complete Table 1.1 to show the number of σ and π bonds in a molecule of N_2 and to describe how the orbitals overlap to form σ and π bonds.

Table 1.1

	σ bond	π bond
number of bonds in N_2		
how the orbitals overlap		

[4]

24. M/J 24/P23/Q4/a**4 (a)** $\text{NH}_3(\text{g})$ reacts with $\text{HCl}(\text{g})$ to produce $\text{NH}_4\text{Cl}(\text{s})$, as shown.

Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl .

[2]

25. MAR 24/P22/Q1/a

1 Bismuth is an element in Group 15 of the Periodic Table.

(a) Bismuth has metallic bonding.

Draw a labelled diagram to show the metallic bonding in bismuth.

[1]

26. MAR 24/P22/Q2/c(i)

2 Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

(c) An excess of Cl_2 reacts with phosphorus to form PCl_5 .

(i) PCl_5 is a simple molecule in the gas phase.

It also exists in a solid form as two ions, PCl_4^+ and PCl_6^- .

Complete Table 2.2 to identify the shapes of each of these species.

Table 2.2

species	PCl_5	PCl_4^+	PCl_6^-
shape		tetrahedral	

[2]

27. MAR 24/P22/Q3/d

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

(d) The N_2F_2 molecule has a double covalent bond between its nitrogen atoms. This consists of a σ and a π bond.

(i) Complete Fig. 3.2 to show the dot-and-cross diagram for N_2F_2 .

Show outer electrons only.

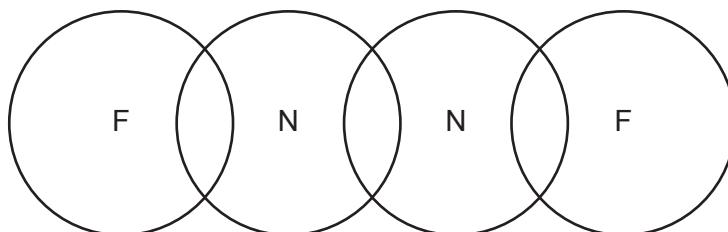


Fig. 3.2

[2]

(ii) Deduce the hybridisation of the N atoms in N_2F_2 .

[1]

(iii) Draw a diagram of the π bond between the N atoms in N_2F_2 and describe how it forms.

[2]

28. O/N 23/P21/Q1/d(iii)

- 1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (d) POC_l₃ shows similar chemical properties to PC_l₅.

POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

- (iii) POC_l₃ contains a double covalent bond between P and O.

Complete the dot-and-cross diagram, in Fig. 1.1, to show the bonding in POC_l₃.

Show outer shell electrons only.

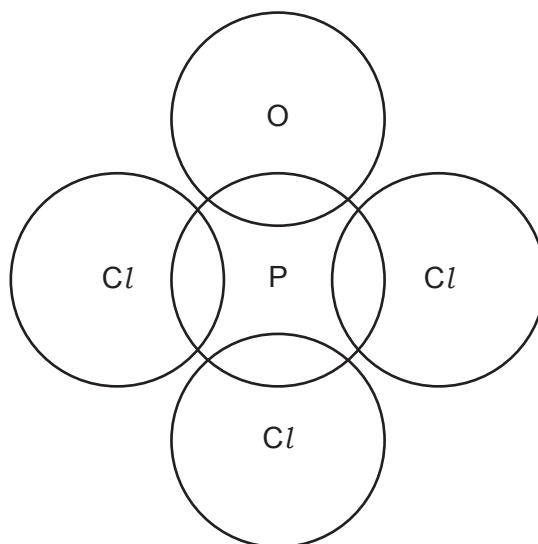
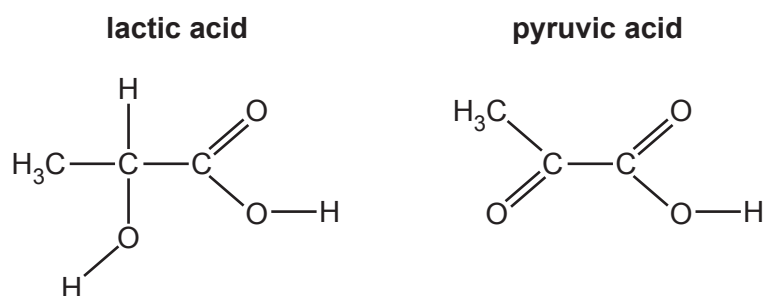


Fig. 1.1

29. O/N 23/P22/Q4/a(iii)

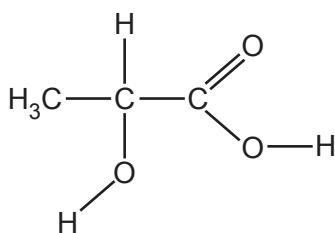
- 4 Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, CH_3COCOOH , both contain two functional groups.

**Fig. 4.1**

- (iii) Lactic acid forms hydrogen bonds with water.

Complete Fig. 4.2 to show the formation of a hydrogen bond between one molecule of lactic acid and one molecule of water.

Label the hydrogen bond. Show any relevant dipoles and lone pairs of electrons.

**Fig. 4.2**

[3]

30. O/N 23/P23/Q1/d(iii)

- 1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (d) POC_l₃ shows similar chemical properties to PC_l₅.

POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

- (iii) POC_l₃ contains a double covalent bond between P and O.

Complete the dot-and-cross diagram, in Fig. 1.1, to show the bonding in POC_l₃.

Show outer shell electrons only.

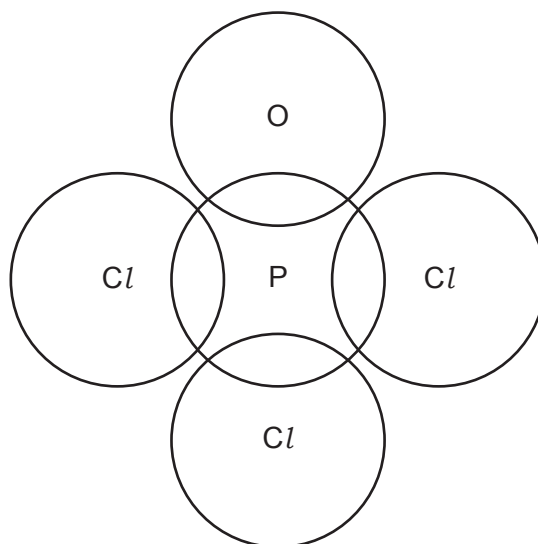


Fig. 1.1

31. M/J 23/P21/Q1/e

- 1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.
- (e) Te reacts with F_2 at 150°C to form TeF_x . Molecules of TeF_x are octahedral with bond angles of 90° .

Explain why TeF_x is octahedral with bond angles of 90° .

[2]

32. M/J 23/P22/Q1/b

- 1 The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point/K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

- (b) (i) Define electronegativity.

[1]

- (ii) Explain why electronegativity increases across a period.

[2]

- (iii) Name the strongest intermolecular force that exists between $\text{NH}_3(\text{l})$ molecules.

[1]

- (iv) Draw a diagram to show the formation of the strongest intermolecular force between **two** molecules of $\text{NH}_3(\text{l})$.

Include any relevant lone pairs of electrons and dipoles.

[2]

- (v) The melting points of ice and ammonia are shown in Table 1.2.

Table 1.2

solid	melting point/K
ice	273
ammonia	195

Suggest **two** reasons for the difference in the melting points of ice and ammonia.

[2]

33. M/J 23/P23/Q1/a(ii)

1 Copper is used in electrical equipment. It has a melting point of 1085 °C.

(ii) Draw a labelled diagram to show the bonding present in copper.

[1]

34. MAR 23/P22/Q1/a,b,e(i)

1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy / kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy / kJ mol ⁻¹	—	7300	2350	3390	2260

(a) (i) Define electronegativity.

[1]

(ii) O and S are in Group 16.

Explain the difference in the Pauling electronegativity values of O and S.

[2]

(b) (i) LiH is an ionic compound.

Draw a dot-and-cross diagram of LiH.

Include **all** electrons.

[2]

(ii) Suggest the shape of a molecule of H₂S.

[1]

(e) (i) Complete Table 1.2 by placing a tick (✓) to show which of the compounds have molecules with an overall dipole moment.

Table 1.2

compound	O=C=O	O=S=O	S=C=S	S=C=O
overall dipole moment				

[2]

35. MAR 23/P22/Q2/a

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

- (a) (i) Draw a labelled diagram to show the bonding and structure of the Group 2 metals at room temperature.

[2]

- (ii) Explain why Mg has a higher electrical conductivity than Na.

[1]

36. MAR 23/P22/Q3/d(iv)

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

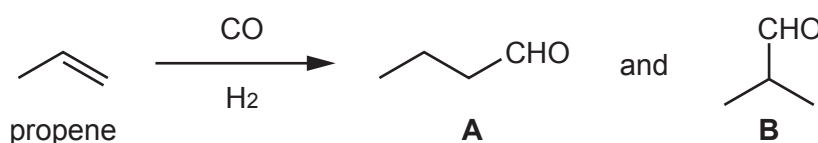


Fig. 3.1

- (d) The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

- (iv) The reaction mixture is cooled to collect CH₃CH₂CHO as a liquid.

Identify all types of van der Waals' forces that are present between molecules of CH₃CH₂CHO.

[1]

37. O/N 22/P21/Q2/a

2 Magnesium shows reactions typical of a Group 2 metal.

- (a) Draw a labelled diagram to show the bonding in magnesium metal.

[2]

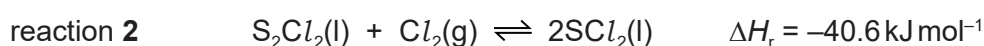
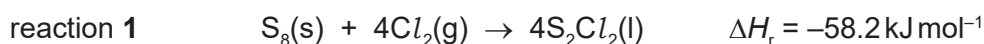
38. O/N 22/P21/Q3/d

also Unit 4 (2 marks), Unit 5 (2 marks), Unit 13 (2 marks), Unit 7 (1 mark); this unit 4 marks

3 Some of the common chlorides of Period 3 elements are shown in the list.



- (d) Sulfur, S₈, reacts with chlorine to form several different chlorides. The most common are S₂Cl₂ and SCl₂. SCl₂ forms when sulfur reacts with an excess of chlorine.



- (i) SCl_2 is a cherry-red liquid that reacts vigorously with water to form an acidic solution.

Use this information to deduce the bonding and structure shown by SCl_2 .
Explain your answer.

[2]

- (ii) Calculate the enthalpy change of formation, ΔH_f , of $\text{SCl}_2(\text{l})$. You may find it useful to use Hess's Law to construct an energy cycle.

enthalpy change of formation of $\text{SCl}_2(\text{l})$, $\Delta H_f =$ kJ mol^{-1}
[2]

- (iii) State the effect of a decrease in pressure on the position of equilibrium in reaction 2.
Explain your answer.

[1]

Fig. 3.1 shows the two structural isomers of S_2Cl_2 .

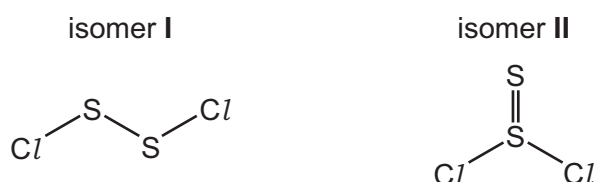


Fig. 3.1

- (iv) Define the term structural isomer.

[2]

- (v) Suggest a value for the Cl-S-S bond angle in isomer I. Explain your answer.

bond angle = $^\circ$

explanation

[2]

- (vi) Draw a dot-and-cross diagram to show the bonding in isomer II. Show outer shell electrons only.

[2]

39. O/N 22/P22/Q1/a,c(iii)

1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

(a) Ionic and covalent bonds both involve an electrostatic attraction between different species.

Identify the species that are electrostatically attracted to one another in:

- an ionic bond
- a covalent bond.

[2]

(c) NH_4^+ is a Brønsted–Lowry acid.

(iii) The nitrogen atom in NH_4^+ is sp^3 hybridised. sp^3 orbitals form from the mixing of one 2s and three 2p orbitals.

Sketch the shapes of a 2s and a $2p_x$ orbital on the axes in Fig. 1.1.

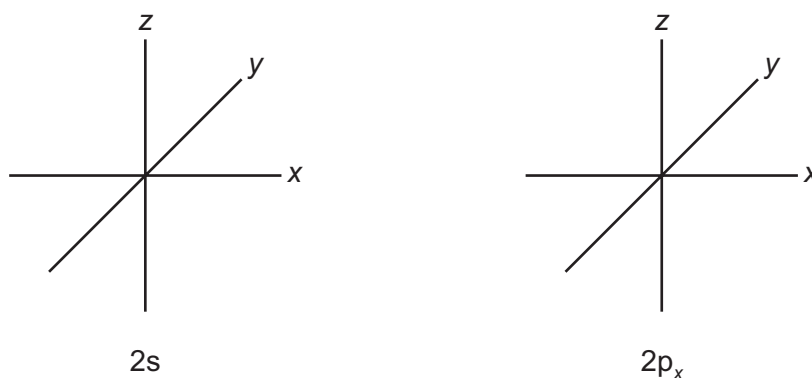


Fig. 1.1

[2]

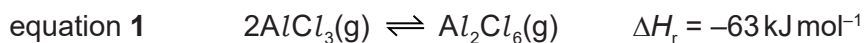
40. O/N 22/P22/Q2/c(i),e(ii),f(i)

2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

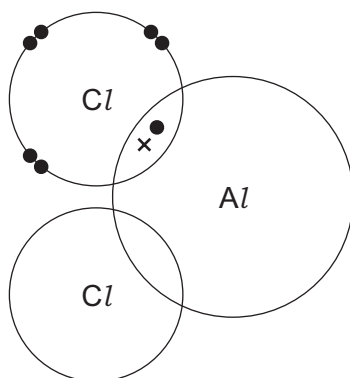
Table 2.1

Period 3 chloride	NaCl	AlCl ₃	SiCl ₄	PCl ₅	PCl ₃	SCl ₂
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

(c) In the gas phase, AlCl₃(g) exists at equilibrium with Al₂Cl₆(g) as shown.



(i) Complete the dot-and-cross diagram to show the bonding in Al₂Cl₆.



[2]

(ii) SCl₂ has $M_r = 103.1$ and is a liquid at room temperature. SBr₂ has $M_r = 191.9$ and is a gas at room temperature.

Explain the difference in the physical state of SCl₂ and SBr₂. Give your answer in terms of intermolecular forces.

[2]

(f) Bismuth is a dense metal in the same group as phosphorus.

(i) Draw a labelled diagram to show the bonding in bismuth metal.

[2]

41. O/N 22/P23/Q2/a

2 Magnesium shows reactions typical of a Group 2 metal.

(a) Draw a labelled diagram to show the bonding in magnesium metal.

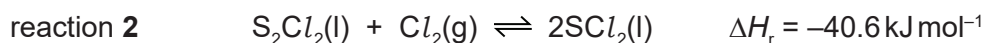
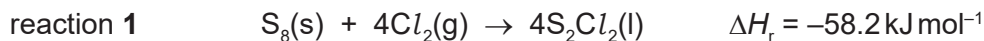
[2]

42. O/N 22/P23/Q3/d

also Unit 4 (2 marks), Unit 5 (2 marks), Unit 13 (2 marks), Unit 7 (1 mark); this unit 4 marks

3 Some of the common chlorides of Period 3 elements are shown in the list.

- (d)** Sulfur, S_8 , reacts with chlorine to form several different chlorides. The most common are S_2Cl_2 and SCl_2 . SCl_2 forms when sulfur reacts with an excess of chlorine.



- (i)** SCl_2 is a cherry-red liquid that reacts vigorously with water to form an acidic solution.

Use this information to deduce the bonding and structure shown by SCl_2 .
Explain your answer.

[2]

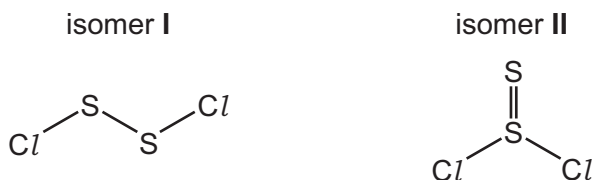
- (ii)** Calculate the enthalpy change of formation, ΔH_{f} , of $\text{SCl}_2(\text{l})$. You may find it useful to use Hess's Law to construct an energy cycle.

enthalpy change of formation of $\text{SCl}_2(\text{l})$, $\Delta H_{\text{f}} =$ kJ mol⁻¹
[2]

- (iii)** State the effect of a decrease in pressure on the position of equilibrium in reaction 2. Explain your answer.

[1]

Fig. 3.1 shows the two structural isomers of S_2Cl_2 .

**Fig. 3.1**

- (iv)** Define the term structural isomer.

[2]

- (v)** Suggest a value for the Cl-S-S bond angle in isomer I. Explain your answer.

bond angle = °

explanation

[2]

- (vi)** Draw a dot-and-cross diagram to show the bonding in isomer II. Show outer shell electrons only.

[2]

43. M/J 22/P21/Q2/a

- 2 Nitrogen molecules, $\text{N}_2(\text{g})$, contain two atoms attracted to each other by a triple covalent bond.

(a) Describe how the triple covalent bond forms in a $\text{N}_2(\text{g})$ molecule. Refer to orbital overlap and hybridisation in your answer.

[3]

44. M/J 22/P22/Q2/b

- 2 Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, RaCl_2 , has a melting point of 900°C and is soluble in water.

(b) Draw a dot-and-cross diagram to show the arrangement of outer electrons in RaCl_2 .

[1]

45. M/J 22/P23/Q3/a(i)

- 3 G belongs to a group of compounds called ethers.

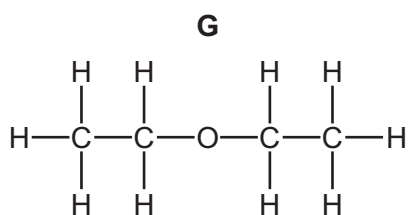
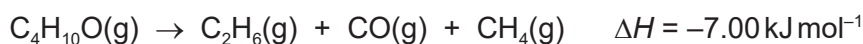


Fig. 3.1

- (a) When **G**, $\text{C}_4\text{H}_{10}\text{O}$, is heated, thermal decomposition occurs.



- (i) The atoms in a molecule of CO are held together by a triple covalent bond. One of these bonds is a coordinate (dative covalent) bond.

Draw a dot-and-cross diagram to show the arrangement of outer electrons in a CO molecule.

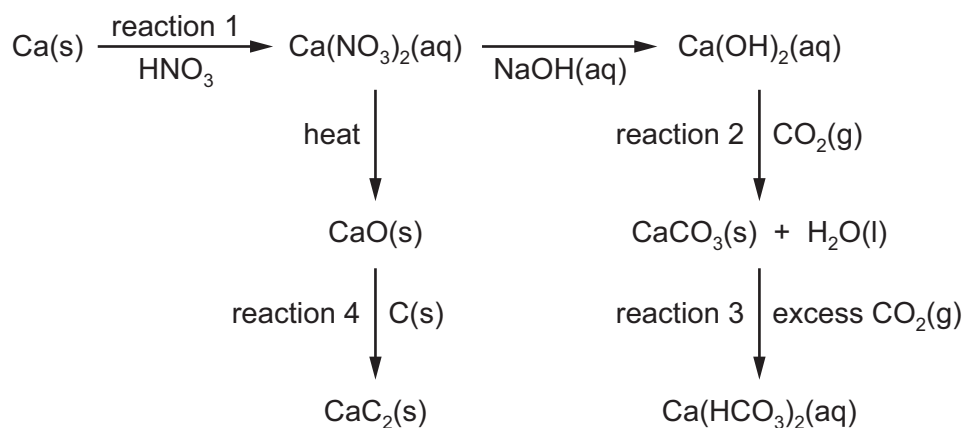
Use • to represent electrons from an oxygen atom.

Use × to represent electrons from a carbon atom.

[1]

46. O/N 21/P21/Q2/c

2 The reaction scheme shows some reactions of calcium.



(c) In reaction 4, calcium carbide, CaC_2 , is formed from CaO .

CaC_2 contains the C_2^{2-} anion. Each carbon in C_2^{2-} is sp hybridised.

(i) Describe how sp hybridised orbitals are formed.

[1]

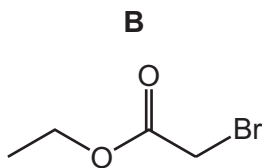
(ii) Sketch a diagram to show how two sp hybrid orbitals can form a sigma (σ) bond.

[2]

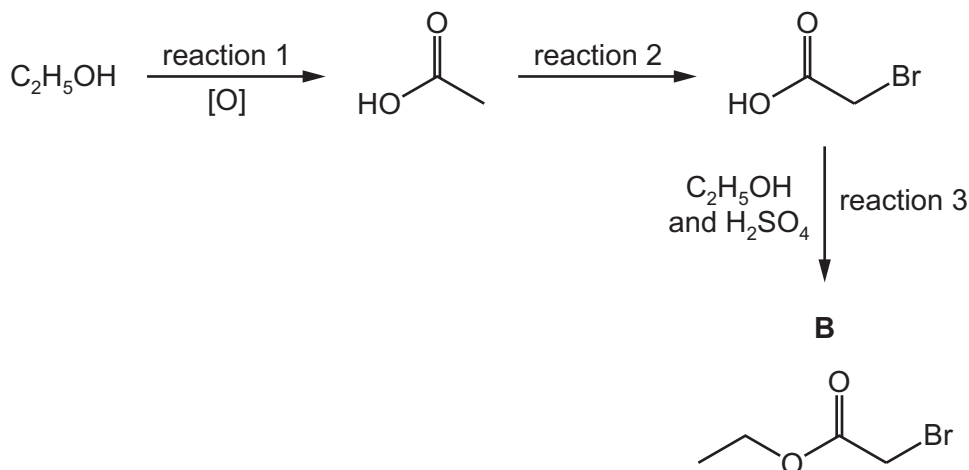
47. O/N 21/P21/Q4/d

Data Booklet needed

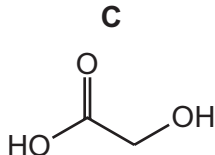
- 4** Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, $\text{C}_2\text{H}_5\text{OH}$.



- (b)** Reaction 2 needs to take place in the absence of water to prevent formation of compound **C**.



If **C** is present in the reaction mixture of reaction 3, a different compound, compound **D**, will also form. Compound **D** has two identical functional groups.

The infrared spectrum of **D** shows strong absorptions at 1100 cm^{-1} and 1720 cm^{-1} , but no absorption due to O–H bonds.

Use the *Data Booklet* to identify the functional group present in **D**.

Explain your answer as fully as you can.

[3]

- (d)** Explain why **C** is very soluble in water.

[1]

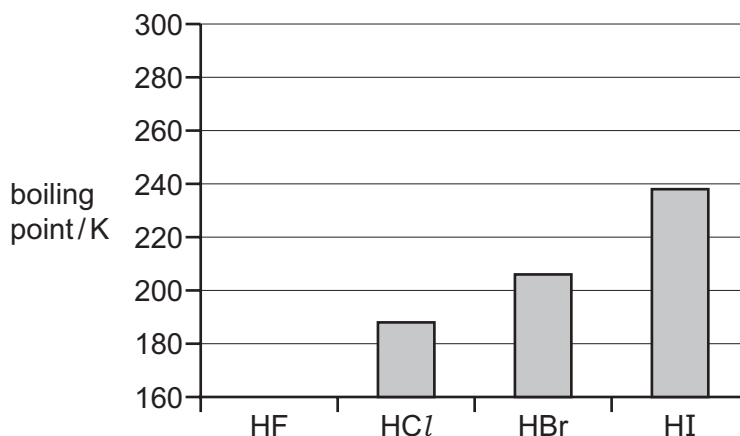
48. O/N 21/P22/Q1/a,f(i)

1 Hydrogen iodide, HI, is a colourless gas at room temperature.

(a) (i) Explain why HI has a higher boiling point than HCl and HBr.

[2]

(ii) The bar chart shows the boiling points of HCl, HBr and HI. The boiling point of HF is not shown.



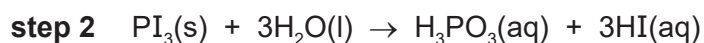
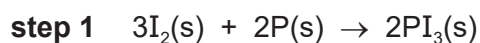
Hydrogen bonds form between HF molecules.

Draw a bar on the bar chart to predict the boiling point of HF.

Explain your answer.

[2]

(f) In the laboratory, HI(aq) can be formed in a two-step process.



(i) Draw a 'dot-and-cross' diagram of a PI_3 molecule.

[2]

49. O/N 21/P22/Q2/d(iii)

(d) Indium and aluminium are elements in Group 13 of the Periodic Table.

Indium has very similar chemical properties to aluminium.

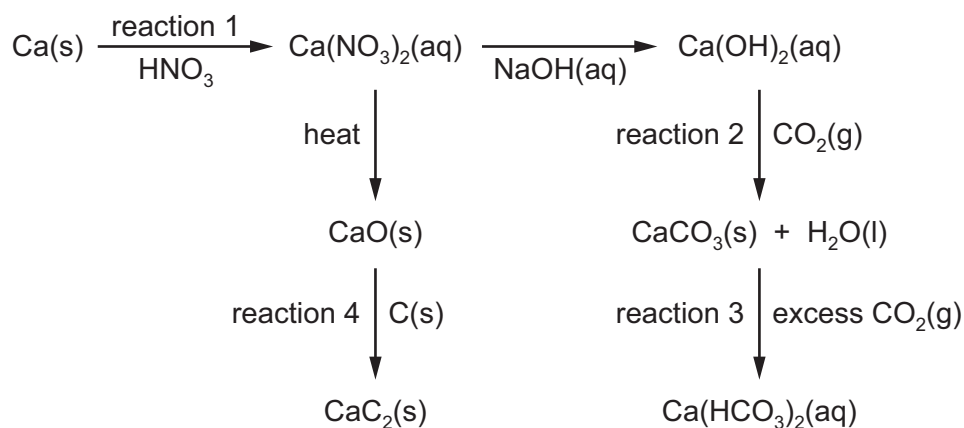
- Indium reacts vigorously with hydrochloric acid to form a colourless gas and a salt in solution.
- Indium oxide, In_2O_3 , is amphoteric.
- Gaseous indium bromide has the formula In_2Br_6 . This molecule contains coordinate bonds.

(iii) Draw a diagram that clearly shows the types of bond present in $\text{In}_2\text{Br}_6(\text{g})$.

[2]

50. O/N 21/P23/Q2/c

2 The reaction scheme shows some reactions of calcium.



(c) In reaction 4, calcium carbide, CaC_2 , is formed from CaO .

CaC_2 contains the C_2^{2-} anion. Each carbon in C_2^{2-} is sp hybridised.

(i) Describe how sp hybridised orbitals are formed.

[1]

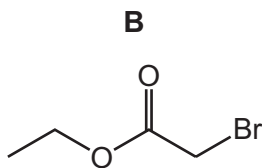
(ii) Sketch a diagram to show how two sp hybrid orbitals can form a sigma (σ) bond.

[2]

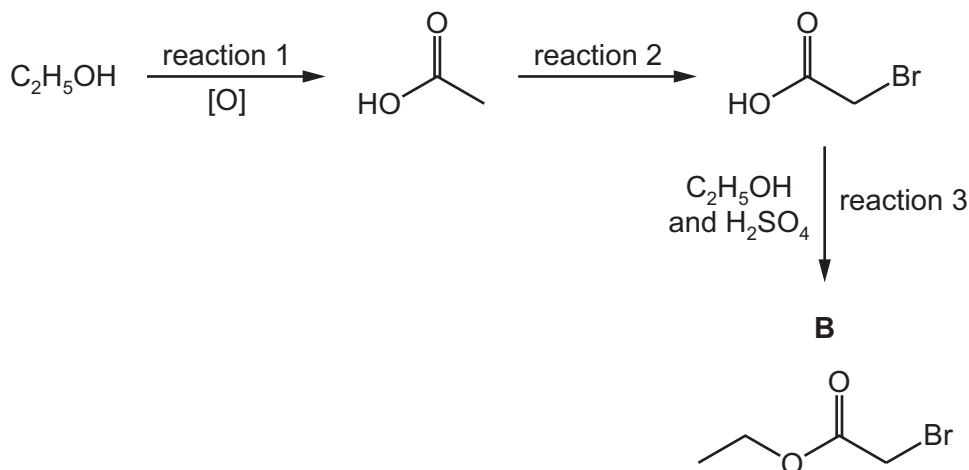
51. O/N 21/P23/Q4/d

Data Booklet needed

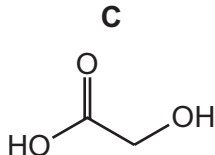
- 4** Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, $\text{C}_2\text{H}_5\text{OH}$.



- (b)** Reaction 2 needs to take place in the absence of water to prevent formation of compound **C**.



If **C** is present in the reaction mixture of reaction 3, a different compound, compound **D**, will also form. Compound **D** has two identical functional groups.

The infrared spectrum of **D** shows strong absorptions at 1100 cm^{-1} and 1720 cm^{-1} , but no absorption due to O–H bonds.

Use the *Data Booklet* to identify the functional group present in **D**.

Explain your answer as fully as you can.

[3]

- (d)** Explain why **C** is very soluble in water.

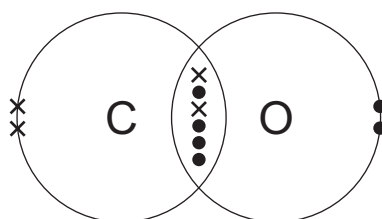
[1]

52. M/J 21/P21/Q2/a,b

Data Booklet needed

2 Carbon monoxide gas, CO(g) , and nitrogen gas, $\text{N}_2(\text{g})$, are both diatomic molecules.

(a) The diagram shows the arrangement of outer electrons in a molecule of CO(g) .



(i) State **one** similarity and **one** difference in the way the atoms in a carbon monoxide molecule are bonded together compared to the atoms in a nitrogen molecule.

[2]

(ii) The table states the electronegativity values of carbon, nitrogen and oxygen atoms.

	C	N	O
electronegativity	2.5	3.0	3.5

Use the electronegativity values and relevant details from the *Data Booklet* to complete the table below.

	N_2	CO
number of electrons per molecule		
type(s) of intermolecular (van der Waals') force		

[2]

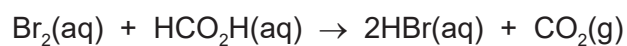
(b) $\text{N}_2(\text{g})$ is less reactive than CO(g) even though $\text{N}_2(\text{g})$ has a lower bond energy than CO(g) .

Suggest why CO(g) is more reactive than $\text{N}_2(\text{g})$.

[1]

53. M/J 21/P21/Q4/d

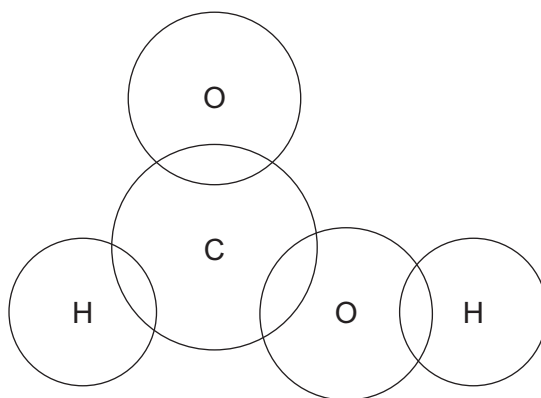
- 4** Aqueous bromine reacts with methanoic acid to form hydrogen bromide and carbon dioxide gas.



The table shows the oxidation numbers of bromine and carbon in the species involved in this reaction.

	Br in Br ₂	C in HCO ₂ H	Br in HBr	C in CO ₂
oxidation number	0	+2	-1	+4

- (d)** Complete the 'dot-and-cross' diagram, showing outer electrons only, to show the bonding in methanoic acid, HCO₂H.



[2]

54. M/J 21/P22/Q2/a(i),b,c(iii,iv)

- 2** The strength of interaction between particles determines whether the substance is a solid, liquid or gas at room temperature.

(a) Lithium sulfide, Li_2S , is a crystalline solid with a melting point of 938°C . It conducts electricity when it is molten.

(i) Give the formulae of the particles present in solid lithium sulfide.

[1]

(b) Carbon monoxide, CO , is a gas at room temperature and pressure. It contains a coordinate bond.

(i) Explain what is meant by *coordinate bond*.

[1]

(ii) Draw a 'dot-and-cross' diagram to show the arrangement of outer electrons in CO .

Show the electrons belonging to the C atom as $\times$.

Show the electrons belonging to the O atom as $\bullet$.

[2]

(c) Nitrogen, N_2 , is also a gas at room temperature and pressure. Neither CO nor N_2 is an ideal gas.

(iii) Complete the table by naming **all** the types of intermolecular forces (van der Waals') in separate samples of $\text{N}_2(\text{g})$ and $\text{CO}(\text{g})$.

	$\text{N}_2(\text{g})$	$\text{CO}(\text{g})$
number of electrons per molecule	14	14
presence of a dipole moment	$\times$	$\checkmark$
boiling point/ $^\circ\text{C}$	-195.8	-191.5
intermolecular forces (van der Waals')		

[2]

(iv) Suggest why the bond in a molecule of CO contains a dipole moment.

[1]

55. M/J 21/P23/Q2/a

- 2** Methanol, CH_3OH , is soluble in water because it forms hydrogen bonds with water molecules.

(a) Draw a fully labelled diagram to show how a hydrogen bond forms between a water molecule and a methanol molecule.

[3]

56. MAR 21/P22/Q2/e(i)

2 Chlorine, Cl_2 , is a reactive yellow-green gas. It is a strong oxidising agent.

(e) Aluminium reacts with chlorine to form aluminium chloride.

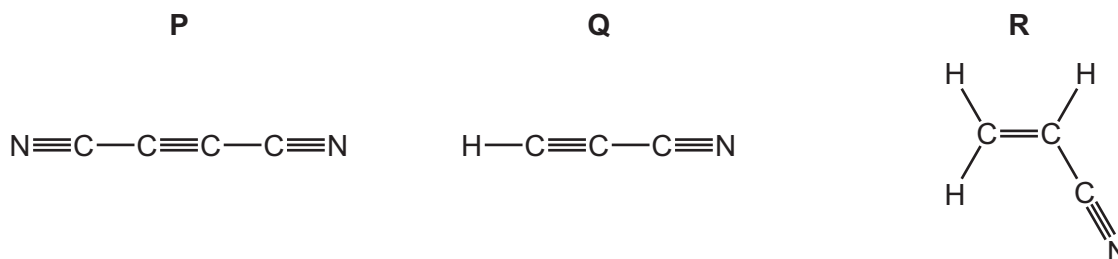
Aluminium chloride can exist as the gaseous molecule $\text{Al}_2\text{Cl}_6(\text{g})$. This molecule contains coordinate bonds.

(i) Draw a diagram that clearly shows all the types of bond present in $\text{Al}_2\text{Cl}_6(\text{g})$.

[2]

57. MAR 21/P22/Q3/b(ii)

3 Compounds **P**, **Q** and **R** have all been found in the atmosphere of one of Saturn's moons.



(b) **Q** forms when HCN reacts with ethyne, $\text{H}-\text{C}\equiv\text{C}-\text{H}$.

(ii) Ethyne, HCN and **Q** all contain triple bonds between two atoms.

A triple bond consists of one sigma (σ) and two pi (π) bonds.

Draw a labelled diagram to show the formation of one pi (π) bond.

[2]

58. O/N 20/P21/Q2/d(i)

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

(d) Chlorine forms several oxides, including Cl_2O , ClO_2 and Cl_2O_6 .

(i) Draw a 'dot-and-cross' diagram of Cl_2O . Show outer-shell electrons only.

[1]

59. O/N 20/P21/Q3/a(i,ii,iv)

- 3** The reducing agent LiAlH_4 can be synthesised by reacting aluminium chloride with lithium hydride, LiH .

(a) (i) At 200°C , aluminium chloride exists as $\text{Al}_2\text{Cl}_6(\text{g})$.

Draw the structure of $\text{Al}_2\text{Cl}_6(\text{g})$, showing fully any coordinate (dative covalent) bonds in the molecule.

[2]

(ii) At 1000°C , aluminium chloride exists as $\text{AlCl}_3(\text{g})$.

State the bond angle in $\text{AlCl}_3(\text{g})$.

° [1]

(iv) LiAlH_4 decomposes slowly to form $\text{LiAl}(\text{s})$ and $\text{H}_2(\text{g})$.



$\text{LiAl}(\text{s})$ shows metallic bonding.

Describe metallic bonding.

[1]

60. O/N 20/P21/Q4/a(iii)

- 4** Iodine is used in many inorganic and organic reactions.

(iii) The table shows the electronegativity values for hydrogen, fluorine and iodine.

element	electronegativity value
H	2.1
F	4.0
I	2.5

Explain, in terms of intermolecular forces, why HI has a lower boiling point than HF .

[2]

61. O/N 20/P22/Q3/d(i)

- 3** PCl_5 , PCl_3 and NCl_3 are halides of Group 15 elements.

(d) NCl_3 is a yellow liquid that can be used to bleach flour.

(i) Predict the shape of the NCl_3 molecule and the Cl-N-Cl bond angle.

shape

bond angle

[2]

62. O/N 20/P23/Q2/d(i)

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

(d) Chlorine forms several oxides, including Cl_2O , ClO_2 and Cl_2O_6 .

(i) Draw a 'dot-and-cross' diagram of Cl_2O . Show outer-shell electrons only.

[1]

63. O/N 20/P23/Q3/a(i,ii,iv)

3 The reducing agent LiAlH_4 can be synthesised by reacting aluminium chloride with lithium hydride, LiH .

(a) (i) At 200°C , aluminium chloride exists as $\text{Al}_2\text{Cl}_6(\text{g})$.

Draw the structure of $\text{Al}_2\text{Cl}_6(\text{g})$, showing fully any coordinate (dative covalent) bonds in the molecule.

[2]

(ii) At 1000°C , aluminium chloride exists as $\text{AlCl}_3(\text{g})$.

State the bond angle in $\text{AlCl}_3(\text{g})$.

° [1]

(iv) LiAlH_4 decomposes slowly to form $\text{LiAl}(\text{s})$ and $\text{H}_2(\text{g})$.



$\text{LiAl}(\text{s})$ shows metallic bonding.

Describe metallic bonding.

[1]

64. O/N 20/P23/Q4/a(iii)

4 Iodine is used in many inorganic and organic reactions.

(iii) The table shows the electronegativity values for hydrogen, fluorine and iodine.

element	electronegativity value
H	2.1
F	4.0
I	2.5

Explain, in terms of intermolecular forces, why HI has a lower boiling point than HF .

[2]

65. M/J 20/P21/Q1/b(ii)

1 Gallium is a metal in Group 13 of the Periodic Table.

(b) Gallium and its compounds show similar properties to aluminium and its compounds.
Gallium reacts with excess chlorine to form gallium trichloride.

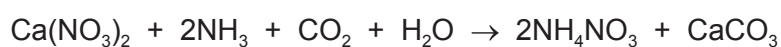
(ii) When gallium trichloride is cooled a solid, Ga_2Cl_6 , forms.

Suggest the name of the attraction formed between two galliumtrichloride molecules to form Ga_2Cl_6 .

[1]

66. M/J 20/P21/Q4/c

4 Calcium nitrate, $\text{Ca}(\text{NO}_3)_2$, reacts with ammonia, carbon dioxide and water to form a mixture of ammonium nitrate and calcium carbonate.



(c) Complete the table to name the shape and give the bond angle of each species.

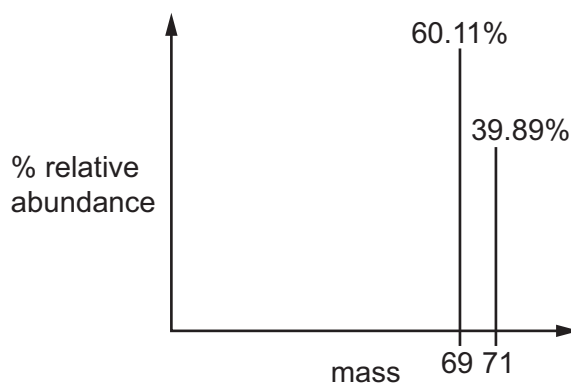
	name of shape	bond angle / °
CO_2		
NH_3		
H_2O		

[3]

67. M/J 20/P22/Q1/d

- 1 Gallium is an element in Group 13.

A sample of gallium is analysed using a mass spectrometer. The mass spectrum produced is shown.



- (d) When gallium is heated in excess chlorine, gallium trichloride, GaCl_3 , is made.

Draw the shape of the gallium trichloride molecule and suggest the Cl-Ga-Cl bond angle.

shape of molecule

bond angle

[2]

68. MAR 20/P22/Q1/d

- 1 Group 2 metals form alkaline solutions in water.

- (d) Beryllium oxide reacts with hydrochloric acid to form molecules of BeCl_2 .

Deduce the bond angle in BeCl_2 .

[1]

69. MAR 20/P22/Q2/d(i)

- 2 The Group 17 elements, chlorine, bromine and iodine, are non-metals that show trends in their physical and chemical properties.

- (d) The hydrogen halides are useful reagents in organic and inorganic reactions.

- (i) Describe and explain the trend in the boiling points of the hydrogen halides, HCl , HBr and HI .

[2]

70. O/N 19/P21/Q2/c(i)

- (c) Selenium is a Group 16 element which shows similar chemical reactions to sulfur.

- (i) Selenium reacts with fluorine to form SeF_6 molecules.

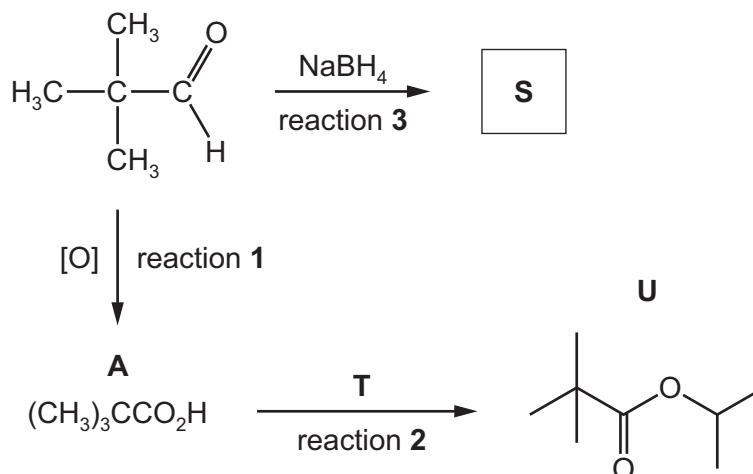
Predict the shape of a molecule of SeF_6 .

[1]

71. O/N 19/P21/Q4/b(ii)

4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(b) Two reaction sequences are shown.



(ii) **A**, $(\text{CH}_3)_3\text{CCO}_2\text{H}$, is a solid at room temperature.

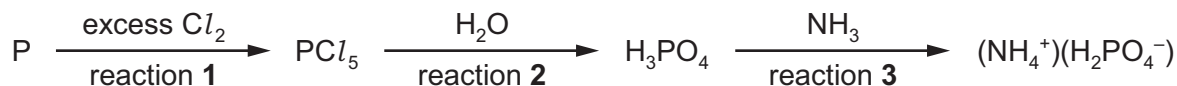
B, $\text{CH}_3\text{CO}_2(\text{CH}_2)_2\text{CH}_3$, is an isomer of **A**. **B** is a liquid at room temperature.

Explain the difference in the physical states of **A** and **B**, with reference to any intermolecular forces that may exist.

[3]

72. O/N 19/P22/Q3/b(ii)

3 A series of reactions for phosphorus and its compounds is shown.



(b) Reaction 3 is a neutralisation reaction in which NH_3 acts as a base.

(ii) Draw the three-dimensional shape of the ammonium ion, NH_4^+ . Give the bond angle.

bond angle = $^\circ$ [1]

73. O/N 19/P23/Q2/c(i)

(c) Selenium is a Group 16 element which shows similar chemical reactions to sulfur.

(i) Selenium reacts with fluorine to form SeF_6 molecules.

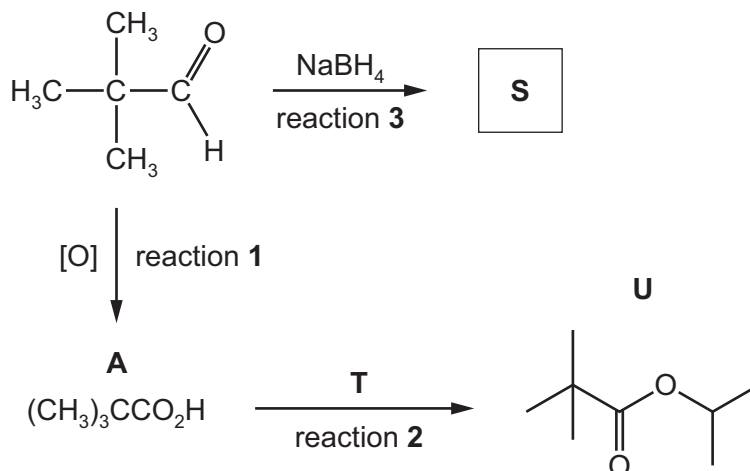
Predict the shape of a molecule of SeF_6 .

[1]

74. O/N 19/P23/Q4/b(ii)

4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(b) Two reaction sequences are shown.



(ii) **A**, $(\text{CH}_3)_3\text{CCO}_2\text{H}$, is a solid at room temperature.

B, $\text{CH}_3\text{CO}_2(\text{CH}_2)_2\text{CH}_3$, is an isomer of **A**. **B** is a liquid at room temperature.

Explain the difference in the physical states of **A** and **B**, with reference to any intermolecular forces that may exist.

[3]

75. M/J 19/P21/Q2/a,d

2 Magnesium silicide, Mg_2Si , is a compound made by heating magnesium with sand.

(a) Draw a 'dot-and-cross' diagram to show the arrangement of outer electrons present in a formula unit of Mg_2Si . Assume magnesium silicide is an ionic compound.

[2]

(d) The table shows the electronegativity values of carbon, hydrogen and silicon.

element	carbon	hydrogen	silicon
electronegativity	2.5	2.1	1.8

(i) C–H and Si–H bonds have weak dipoles.

Use the electronegativity values in the table to show the polarity of the C–H and Si–H bonds.



[2]

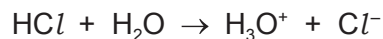
(ii) Explain why methane, CH_4 , has no overall dipole moment.

[2]

76. M/J 19/P22/Q2/c(iii,iv)

2 Group 17 elements are commonly referred to as the halogens.

(c) Hydrogen chloride reacts with water.



(iii) Name the type of bond formed between H^+ and H_2O to make H_3O^+ .

[1]

(iv) For H_3O^+ , predict its shape and the H–O–H bond angle.

shape

bond angle

[2]

77. M/J 19/P22/Q3/a(iv)

3 Period 3 elements react with chlorine gas, $\text{Cl}_2(\text{g})$, to form chlorides.

(a) The table shows the differences in observations which occur when two Period 3 chlorides are added to water.

Period 3 chloride	observations when added to water	pH of solution formed with water
NaCl	White solid disappears. Colourless solution made.	7
SiCl_4	Pale yellow solution forms. Bubbles form and the test-tube feels hot. White precipitate forms.	1–2

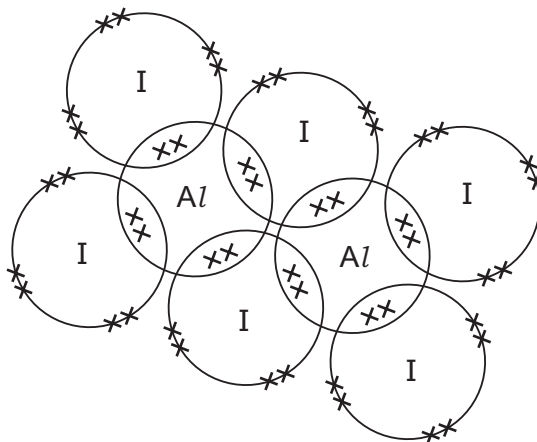
(iv) Explain, in terms of electronegativity, why the bonding in NaCl is different from the bonding in SiCl_4 .

[3]

78. M/J 19/P23/Q2/b

2 Iodine is an element in Group 17 of the Periodic Table.

(b) The diagram shows the arrangement of the outer electrons within a molecule of Al_2I_6 .



(i) How many co-ordinate (dative covalent) bonds are made when a molecule of Al_2I_6 is formed from its atoms?

[1]

(ii) Describe how co-ordinate (dative covalent) bonds form within this molecule.

[1]

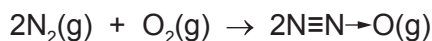
79. MAR 19/P22/Q1/d(iii)

Data Booklet needed

1 Nitrogen, N_2 , is the most abundant gas in the Earth's atmosphere and is very unreactive.

(d) Three oxides of nitrogen, NO , NO_2 and N_2O , can be formed under different conditions.

(iii) Molecules of N_2O can be formed by the reaction between N_2 and O_2 . The bond between the N and O atoms ($N \rightarrow O$) is a co-ordinate (dative covalent) bond.



The enthalpy change of reaction for this reaction is $+82 \text{ kJ mol}^{-1}$.

Calculate the bond enthalpy, in kJ mol^{-1} , of the $N \rightarrow O$ bond.

Use relevant data from the *Data Booklet* to answer this question.

bond enthalpy of the $N \rightarrow O$ bond = kJ mol^{-1}
[2]

80. MAR 19/P22/Q2/a(ii),c(i,iv)

- 2 The elements in Group 17 of the Periodic Table are called the halogens. They form stable compounds with both metals and non-metals.

The table gives some data about F_2 , HCl and CaF_2 .

	F_2	HCl	CaF_2
boiling point/K	85	188	2773
relative formula mass	38.0	36.5	78.1

- (ii) F_2 and HCl are both covalent molecules.

Suggest why the boiling point of HCl is higher than that of F_2 .

[2]

- (c) HO is the only known molecule that contains only the elements hydrogen, oxygen and fluorine.

- (i) Draw a 'dot-and-cross' diagram to represent the bonding in a molecule of HO .

Show the outer shell electrons only.

[2]

- (iv) Pure HF is a colourless liquid at 273 K. The liquid contains HF molecules that have strong hydrogen bonds between them.

Draw a fully labelled diagram to suggest how a hydrogen bond can form between two HF molecules.

[3]

81. O/N 18/P21/Q1/b

- 1 Iron pyrite, FeS_2 , has a yellow colour that makes it look like gold metal. The compound contains the ions Fe^{2+} and S_2^{2-} .

- (b) Describe the metallic bonding in gold.

[2]

82. O/N 18/P21/Q2/a(ii)

- 2 (a) Nitrogen, N_2 , is an inert gas that makes up 78% of the Earth's atmosphere.

- (ii) Draw a 'dot-and-cross' diagram of a nitrogen molecule. Show outer electrons only.

[1]

83. O/N 18/P22/Q1/b

- 1 The model of the nuclear atom was first proposed by Ernest Rutherford. He developed this model on the basis of results obtained from an experiment using gold metal foil.

- (b) State the type of bonding in gold.

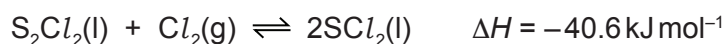
[1]

84. O/N 18/P22/Q2/b(i)

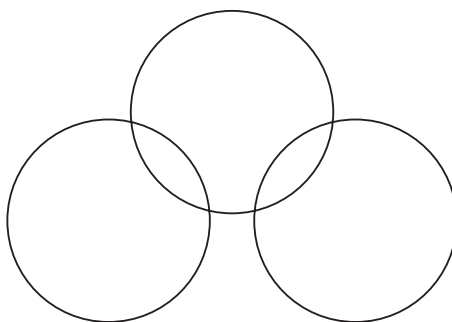
2 The table gives some data for elements in the third period and some of their compounds.

element	Na	Mg	Al	Si	P	S
type of bonding	metallic				covalent	covalent
formula of oxide					P ₄ O ₁₀	SO ₂
formula of chloride	NaCl	MgCl ₂				SCl ₂

(b) SCl₂ is formed in the following reaction.



(i) Complete the 'dot-and-cross' diagram to show the bonding in a molecule of SCl₂. Show outer electrons only.



[1]

85. O/N 18/P23/Q1/b

1 Iron pyrite, FeS₂, has a yellow colour that makes it look like gold metal. The compound contains the ions Fe²⁺ and S₂²⁻.

(b) Describe the metallic bonding in gold.

[2]

86. O/N 18/P23/Q2/a(ii)

2 (a) Nitrogen, N₂, is an inert gas that makes up 78% of the Earth's atmosphere.

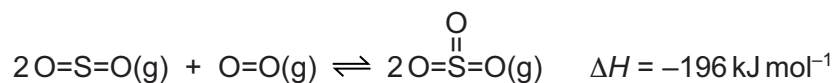
(ii) Draw a 'dot-and-cross' diagram of a nitrogen molecule. Show outer electrons only.

[1]

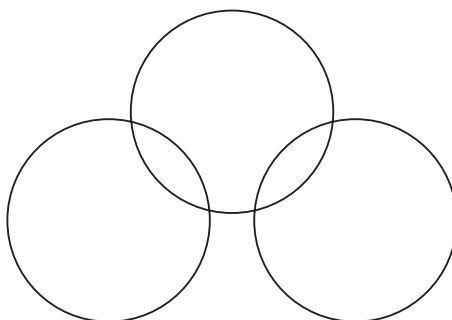
87. M/J 18/P21/Q1/f(i)

- 1 Sulfuric acid is manufactured by the Contact process.

One stage in this process is the conversion of sulfur dioxide into sulfur trioxide in the presence of a heterogeneous catalyst of vanadium(V) oxide, V_2O_5 .



- (f) SO_2 reacts with water to form sulfurous acid.
Sulfurous acid is a weak Brønsted-Lowry acid, while sulfuric acid is a strong Brønsted-Lowry acid.
- (i) Complete the 'dot-and-cross' diagram to show the bonding in a molecule of SO_2 . Show outer electrons only.



[1]

88. M/J 18/P21/Q2/a

- 2 Crude oil is a complex mixture of hydrocarbon molecules.

The hydrocarbon molecules in crude oil are separated by fractional distillation. Fractional distillation is used because the different hydrocarbon molecules in crude oil have different boiling points.

- (a) Explain why the hydrocarbon molecules in crude oil have different boiling points.

[2]

89. M/J 18/P22/Q1/b(iv)

- 1 Sodium and magnesium are the first two elements in the third period.

(b) Group 2 elements, including magnesium, react with oxygen and with dilute acids. There are trends in both the physical and chemical properties of the elements and their compounds down the group. Reactivity generally increases from Mg to Ba.

- (iv) Suggest why there is a general **decrease** in the melting points of the elements down Group 2.

[3]

90. M/J 18/P22/Q2/b

2 Ammonium iron(II) sulfate, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2$, has a relative formula mass, M_r , of 284.

(b) One of the cations in ammonium iron(II) sulfate is the ammonium ion, NH_4^+ .

(i) Draw a 'dot-and-cross' diagram of an ammonium ion. Show outer shell electrons only.

Use $\times$ to show electrons from nitrogen.

Use $\bullet$ to show electrons from hydrogen.

[2]

(ii) Suggest the shape of an ammonium ion and predict the bond angle.

shape

bond angle

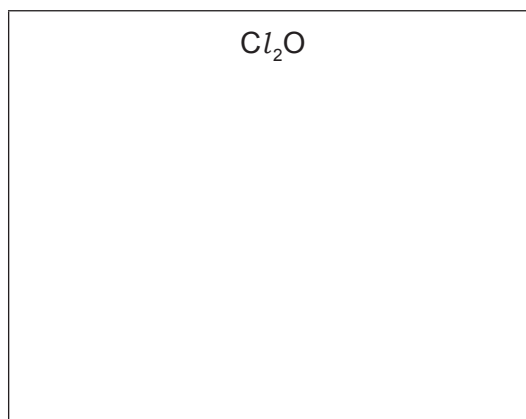
[2]

91. M/J 18/P23/Q1/a

1 The elements sodium to chlorine, in the third period, all form oxides.

(a) Draw a diagram to show the shape of the molecule of each of the oxides, SO_3 and Cl_2O .
Name each shape.

In SO_3 each oxygen atom forms a double bond with the sulfur atom.



[4]

92. MAR 18/P22/Q2/d(iv)

2 Carbon and silicon are elements in Group 14.

(d) Silicon shows the same kind of bonding and structure as diamond.

(iv) Predict the shape of the SiH_4 molecule.

[1]

93. MAR 18/P22/Q3/b(i)

3 Calcium and its compounds have a large variety of applications.

(b) Calcium ethanedioate is formed when calcium reacts with ethanedioic acid, $(\text{CO}_2\text{H})_2$. The compound contains one cation and one anion.

(i) Draw the 'dot-and-cross' diagram of the cation present in calcium ethanedioate. Show **all** electrons.

[1]

94. O/N 17/P22/Q1/a,b(ii,iii)

1 The elements sodium to sulfur react with chlorine. The melting points of some of the chlorides formed are shown.

chloride	NaCl	MgCl_2	AlCl_3	SiCl_4	PCl_3	SCl_2
melting point/K	1074	987	463	203	161	195

(a) Predict the shapes of AlCl_3 and PCl_3 .

Draw diagrams to show the shapes, name the shapes and state the bond angles.

AlCl_3	PCl_3
shape angle	shape angle

[4]

(ii) Explain why the melting point of SiCl_4 is higher than that of PCl_3 .

[2]

(iii) Draw the 'dot-and-cross' diagram of a molecule of SiCl_4 .
Show outer electrons only.

[1]

95. O/N 17/P22/Q3/b(ii)

- 3** The elements in Group 2 show trends in their properties that are typical of metals.
The elements in Group 17 show trends in their properties that are typical of non-metals.

(ii) The melting point decreases down Group 2.

Explain this trend.

[2]

96. M/J 17/P21/Q2/b

- 2** Structure and bonding can be used to explain many of the properties of substances.

(b) (i) Name the strongest type of intermolecular force in ice.

[1]

(ii) Draw a fully labelled diagram of two water molecules in ice, showing the force in (i) and how it forms.

[3]

97. M/J 17/P22/Q4/a(ii)

- 4** (a) The hydrocarbons **A**, C_4H_{10} , and **B**, C_4H_8 , are both unbranched.

A does **not** decolourise bromine.

B decolourises bromine and shows geometrical isomerism.

(ii) The hydrocarbon **A**, C_4H_{10} , has a branched isomer.

Suggest why unbranched **A** has a higher boiling point than its branched isomer.

[2]

98. MAR 17/P22/Q1/c(i)

(c) Cerium is a lanthanoid metal that shows similar chemical reactions to some elements in the third period. Most of cerium's compounds contain Ce^{3+} or Ce^{4+} ions.

(i) Cerium shows the same structure and bonding as a typical metal.

Draw a labelled diagram to show the structure and bonding in cerium.

[2]

99. MAR 17/P22/Q2/a(i,ii)

- 2 Hydrogen halides are compounds formed when halogens (Group 17 elements) react with hydrogen. The bond polarity of the hydrogen halides decreases from HF to HI.

Some relevant data are shown in the table.

hydrogen halide	HF	HCl	HBr	HI
boiling point/°C	19	−85	−67	−35
H–X bond energy/kJ mol ^{−1}	562	431	366	299

- (a) (i) Explain the meaning of the term *bond polarity*.

[1]

- (ii) Suggest why the boiling point of HF is **much** higher than the boiling points of the other hydrogen halides.

[2]

100. M/J 16/P22/Q2/b

- 2 The elements in Group 17, the halogens, and their compounds, show many similarities and trends in their properties. Some data are given for the elements fluorine to iodine.

element	bond energy /kJ mol ^{−1}	standard enthalpy change of atomisation, $\Delta H_{\text{at}}^{\ominus}$ /kJ mol ^{−1}	boiling point of element /K	boiling point of hydrogen halide /K
fluorine, F–F	158	79	85	293
chlorine, Cl–Cl	242	121	238	188
bromine, Br–Br	193	112	332	206
iodine, I–I	151	107	457	238

- (b) (i) Explain the trend in the boiling points of the hydrogen halides, HCl, HBr and HI.

[2]

- (ii) Suggest why the hydrogen halide HF does not follow the trend in boiling points shown by HCl, HBr and HI.

[2]

101. M/J 16/P23/Q1/b(i)

1 An experiment was carried out to determine the percentage of iron in a sample of iron wire.

(b) Some electronegativity values are shown.

element	electronegativity
aluminium	1.5
chlorine	3.0
iron	1.8

(i) Use the data to suggest the nature of the bonding in iron(III) chloride. Explain your answer.

[2]

102. M/J 16/P23/Q2/d

2 Ammonium nitrate is an important fertiliser made by the acid-base reaction between ammonia and nitric acid.

(d) (i) Draw a dot-and-cross diagram of the ammonium ion. Show the outer electrons only. Use the following code for your electrons.

- electrons from nitrogen
- × electrons from hydrogen

[2]

(ii) State the shape of an ammonium ion and give the H–N–H bond angle.

shape

bond angle

[2]

103. MAR 16/P22/Q1/a(iv)

1 This question is about Period 3 elements and their compounds.

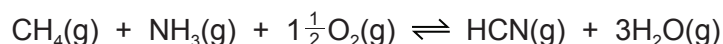
(a) Give an explanation for each of the following statements.

(iv) Magnesium is a better electrical conductor than sodium.

[1]

104. MAR 16/P22/Q3/a

- 3 Over one million tonnes of hydrogen cyanide, HCN, are produced each year using the Andrussow process. The overall equation for the reaction is shown.



- (a) (i) Draw a dot-and-cross diagram to represent the bonding in a molecule of ammonia, NH_3 , and state the shape of the molecule.

shape of molecule

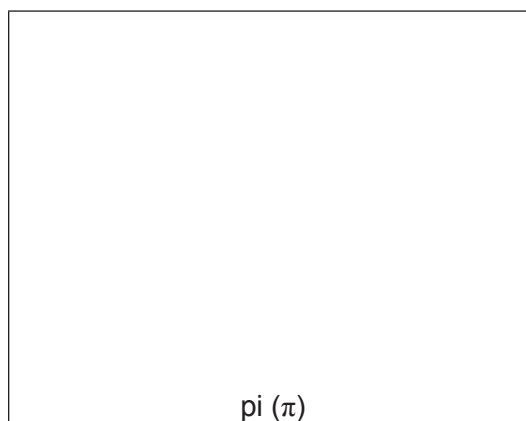
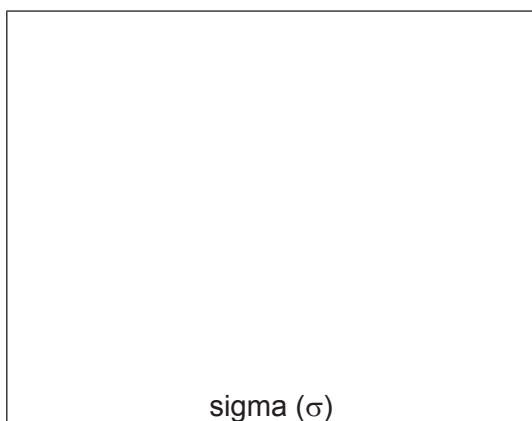
[3]

- (ii) A molecule of hydrogen cyanide, HCN, is shown.



The bonding between the carbon and nitrogen atoms consists of one sigma (σ) bond and two pi (π) bonds.

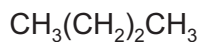
Sketch the shape of the sigma bond and one of the pi bonds in the space below. Show clearly the position of the atomic nuclei in each diagram.



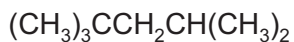
[2]

105. MAR 16/P22/Q4/d

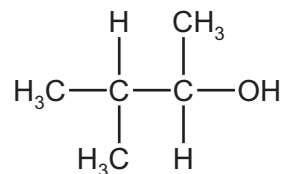
- 4 The following compounds were all found to be components of a sample of petrol.



G



H



J

- (d) The boiling points of compounds **G**, **H** and **J** are shown below.

compound	G	H	J
boiling point/ $^{\circ}\text{C}$	0	99	112

Explain the differences in the boiling points of the three compounds.

[4]

106. O/N 15/P22/Q1/b,c

(b) One of the factors that determines the type of bonding present between the particles of a substance is the relative electronegativities of the bonded particles.

(i) Explain the meaning of the term *electronegativity*.

[2]

(ii) Name and describe the type of bonding you would expect to find between particles with equal electronegativities.

[2]

(iii) Name and describe the type of bonding you would expect to find between particles with very different electronegativities.

[2]

(c) The boiling points of some molecules with equal numbers of electrons are given.

substance	fluorine	argon	hydrogen chloride	methanol
formula	F ₂	Ar	HCl	CH ₃ OH
boiling point/K	85	87	188	338

(i) Explain why the boiling points of fluorine and argon are so similar.

[2]

(ii) Explain why the boiling point of hydrogen chloride is higher than that of fluorine.

[2]

(iii) Explain why methanol has the highest boiling point of all these molecules.

[2]

107. M/J 15/P23/Q1/e

1 Neon is a noble gas.

(e) Neon and argon can both be obtained by fractional distillation of liquid air as they have different boiling points.

Neon has a boiling point of 27.3 K. The boiling point of argon is 87.4 K.

(i) Name the force that has to be overcome in order to boil neon or argon and explain what causes it.

[3]

(ii) Explain why argon has a higher boiling point than neon.

[2]

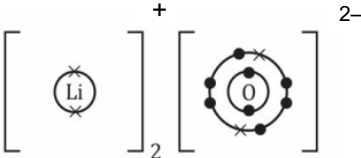
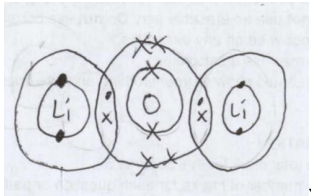
Answers Section

1. M/J 26/P21/Q1/a,c,d

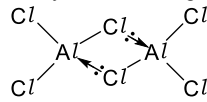
1(a)(i)		1	A any symbol for electrons – symbols can all be the same or different A electrons can be on the lines or in the spaces A non-bonding electrons need not drawn in pairs I inner electrons						
1(a)(ii)	octahedral	1	A square bipyramidal A octahedron A near miss in spelling e.g. octohedral I octagonal / octagon						
1(c)(i)	M1 104.5(°) M2 2 bonding pairs (of electrons) and 2 lone pairs (of electrons) M3 lone pairs (of electrons) repel more than bonding pairs (of electrons)	3	M1 A range 103–105° inclusive I reference to stated shape even if incorrect I reference to tetrahedral geometry and 109.5 in text A 'bonding pairs' and 'lone pairs' assuming they are describing electrons unless stated otherwise. A 2 shared pairs for 2 bonding pairs For 2 lone pairs: A 2 pair of lone pairs A 2 non-bonding pairs I S—Cl bond / 2 bonds for bonding pair I any electrons on outer shell of chlorine R one unpaired electron If M2 is not awarded from text, award M2 for a suitably annotated Figure 1.2 or dot-and-cross diagram The electrons in both bond pairs and lone pairs must be shown on Figure 1.2 M3 A repulsion from lone pair greater than from bond pair M3 A repulsion lone pair – lone pair > lone pair – bonding pair > bonding pair – bonding pair M3 A repulsion lone pair – bonding pair > bonding pair – bonding pair Max 1 for 2 bonding pairs and 1 lone pair AND between 107–109° inclusive						
1(c)(ii)	S as δ+ AND Both Cl as δ–	1	A δ+ δ+ / δ2+ / 2+δ on S A +δ and –δ I Δ I any electrons drawn on the atoms R ± without δ BOD TWO overall dipole direction arrows parallel to bonds I ONE overall dipole direction arrow / arrow						
1(d)(i)	<table><tr><td></td><td>SCl₂</td><td>H₂O</td></tr><tr><td>total number of electrons per molecule</td><td>50</td><td>10</td></tr></table>		SCl ₂	H ₂ O	total number of electrons per molecule	50	10	1	
	SCl ₂	H ₂ O							
total number of electrons per molecule	50	10							
1(d)(ii)	SCl₂ AND greater number of electrons (per molecule) OR H₂O has weaker id (id) as it has smaller number of electrons (per molecule)	1	greater = higher = larger = bigger = more M1 awarded if incorrect number of electrons are given in 1(d)(i) AND still greater than 10 A ecf from 1(d)(i) for SCl₂ < H₂O AND fewer (than 10) electrons per molecule if less than 10 shown in 1(d)(i)) A ecf from 1(d)(i) SCl₂ = H₂O AND same number of electrons per molecule if 10 shown in 1(d)(i) I greater electron density / electron cloud I SCl₂ has more valence / outer electrons						

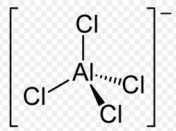
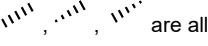
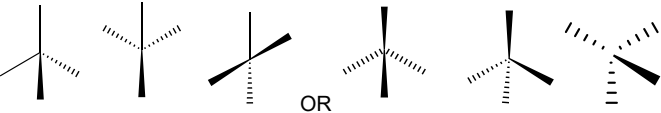
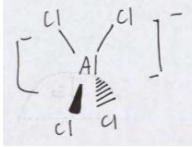
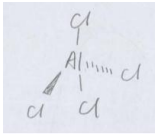
1(d)(iii)	electronegativity difference (between H and O) is greater (than S and Cl)	1 A larger for greater / bigger owtte A H–O has large electronegativity difference and S–Cl has small electronegativity difference I H₂O is more polar than SCl₂ I oxygen/O is more electronegative than chlorine I any comments on hydrogen bonding R H₂O is more electronegative than SCl₂ R O₂ is more electronegative than Cl₂
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2. M/J 26/P22/Q1/b

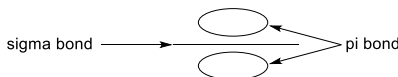
1(b)		2 M1 correct charges M1 A –2 for O ion and +1 / 1+ for Li ion M2 all else fully correct I symbols for Li and O A any symbol for electrons Max 1 for incorrect identification of type of bonding as covalent as long as all else correct  Max 1 for any ionic compound with a positively charged species and a negatively charged species
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3. M/J 26/P24/Q1/c,d

1(c)	Lone pair (of electrons) on a Cl (could be seen as : on a diagram) AND donated / shared with Al (could be seen by the arrow on a diagram) OR A fully annotated Figure 1.2: 	1 A full annotations on Figure 1.2 i.e. lone pair AND dative arrow on <u>both</u> Cl to Al bonds R dative bond shown from Al to Cl on diagram R chloride / chlorine ion BOD long pair A chlorine A donates / gives / shares / provides a lone pair / electron pair / both electrons <u>to Al</u>
1(d)(i)	M1 (name of shape) tetrahedral M2 (bond angle) 109–109.5°	2 M1 A tetrahedra M1 A phonetically spelt name of shape

1(d)(ii)	3D shape required:  Acceptable 3D descriptions of different bond types behind the plane and out of the plane are shown below  are all interchangeable  are all interchangeable Although the 3D structure shown in the syllabus is drawn as the example shown in the mark scheme it is acceptable to allow a combination of bond types shown below (3D structure has two types of wedge adjacent)  A BOD any structures which show unambiguous use of bond shapes representing bonds behind the plane and out of the plane as long as they are different.	1 I negative charge (not required) I bond angles I dative bonds A BOD  R 
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4. M/J 26/P24/Q2/a(i)

2(a)(i)	M1 sideways overlap of (adjacent) p orbitals M2 above and below the (covalent/σ) bond	2 M1 A unambiguous labelled diagram A 'sideways overlap of p' A sideways = lateral = side on = side by side = side to side = parallel I indirect I horizontal / vertical I shoulder-to-shoulder M2 A above and below the (covalent) bond A labelled diagram: 
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5. MAR 26/P22/Q2/a(ii)

2(a)(ii)		1
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6. MAR 26/P22/Q4/b

4(b)	M1 clearly distinct / labelled H bond M2 from lone pair on any oxygen to H atom attached to O M3 with three correct successive partial charges across two molecules	3
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7. O/N 25/P22/Q3/d(ii)

3(d)(ii)	M1 B < A < hexane. M2 hexane / unbranched has stronger / more IMFs M3 hexane / unbranched requires most / more energy	3
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8. O/N 25/P24/Q1/b(v)

1(b)(v)	M1 (trigonal) pyramidal M2 three bonding pairs AND one lone pair (of electrons)	2
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9. M/J 25/P21/Q1/a(ii)

1(a)(ii)	delocalised electrons move (through the structure)	1
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10. M/J 25/P23/Q4/b

4(b)(i)	M1 non-linear M2 bond angle – $120^\circ > x \geq 116^\circ$	2
4(b)(ii)	M1 weaker dipole moment (in SO_2) M2 electronegativity (difference) is larger (/ 1.4) for H and O than for S and O (/ 0.9) OR S=O bond is less polar than H—O OR stronger dipoles in S=O / SO_2	2

11. M/J 25/P24/Q2/a

2(a)(i)	non-linear	1
2(a)(ii)	104.5°	1

12. M/J 25/P24/Q3/d(i)

3(d)(i)	(electrostatic) attraction between the nuclei (of two atoms) and a shared pair of electrons	1
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13. MAR 25/P22/Q1/b(ii)

1(b)(ii)	trigonal bipyramid(al)	1
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14. MAR 25/P22/Q2/a

2(a)	electrostatic attraction between positive metal ions / cations and delocalised electrons	2
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15. O/N 24/P21/Q3/a(iii),d

3(a)(iii)	its delocalised electrons are free to move	1
3(d)(i)	M1 molecules CH_4 to PbH_4 have greater number of electrons / number of electrons increase	1
	M2 greater / stronger instantaneous dipole–induced forces / London forces / dispersion forces OR more energy required to overcome the instantaneous dipole–induced forces / London forces / dispersion forces	1
3(d)(ii)	tetrahedral	1

16. O/N 24/P22/Q2/a(ii),c

2(a)(ii)	Al^{3+} 	1
2(c)(i)	(molecules) H_2S to H_2Te / they have greater number of electrons	1
	stronger instantaneous dipole–induced dipole / London dispersion forces OR more energy required to overcome the instantaneous dipole–induced dipole / London dispersion forces	1

2(c)(ii)	(only) H ₂ O has hydrogen bonding AND hydrogen bonding (much) stronger than the other intermolecular forces OR H ₂ O has hydrogen bonding AND because of higher electronegativity of O compared to S OR stronger van der Waals' forces because (only) H ₂ O has hydrogen bonding OR hydrogen bond AND increased strength of permanent dipoles in H ₂ O outweighs the increase in strength of d-d in the others	1
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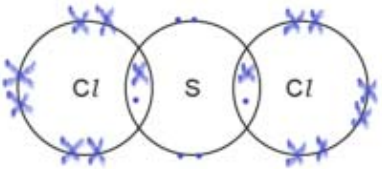
17. O/N 24/P22/Q3/b(iv),f(i)

3(b)(iv)	tetrahedral	1
3(f)(i)	M1 one sigma / σ bond AND head-on (overlap of) p / sp (orbitals)	1
	M2 two pi / π bond(s) AND side-on (overlap of / involving) p (orbitals)	1

18. O/N 24/P23/Q3/a(iii),d

3(a)(iii)	its delocalised electrons are free to move	1
3(d)(i)	M1 molecules CH ₄ to PbH ₄ have greater number of electrons / number of electrons increase	1
	M2 greater / stronger instantaneous dipole-induced forces / London forces / dispersion forces OR more energy required to overcome the instantaneous dipole-induced forces / London forces / dispersion forces	1
3(d)(ii)	tetrahedral	1

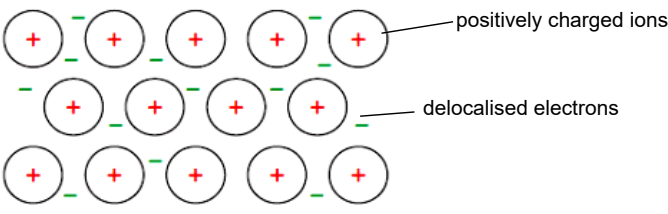
19. M/J 24/P21/Q2/b

2(b)(i)	M1 1 shared pair of electrons between each S-Cl shown as •• M2 6 non-bonding electrons around each Cl shown as • AND 4 non-bonding electrons around S shown as • 	2
2(b)(ii)	M1 shape non-linear M2 bond angle 103–105(°)	2

20. M/J 24/P21/Q5/a

5(a)	(electrostatic) attraction between nuclei of two atoms and shared pair of electrons	1
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21. M/J 24/P22/Q1/c

1(c)	 labelled diagram shows: - regular arrangement of minimum of 2 × 2 circles containing + for positive ions AND surrounded by (sea of) delocalised electrons	1
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22. M/J 24/P22/Q3/d

3(d)	M1 6 electrons between C and N AND 2 electrons between C and S M2 3 lone pairs around S AND none around C AND 1 lone pair around N	2
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23. M/J 24/P23/Q1/a(ii)

1(a)(ii)	<table border="1"> <thead> <tr> <th></th><th>σ bond</th><th>π bond</th></tr> </thead> <tbody> <tr> <td>no of bonds</td><td>1</td><td>2</td></tr> <tr> <td>how orbitals overlap</td><td>direct (orbital overlap)</td><td>sideways (overlap of adjacent p orbitals)</td></tr> </tbody> </table>		σ bond	π bond	no of bonds	1	2	how orbitals overlap	direct (orbital overlap)	sideways (overlap of adjacent p orbitals)	4
	σ bond	π bond									
no of bonds	1	2									
how orbitals overlap	direct (orbital overlap)	sideways (overlap of adjacent p orbitals)									

24. M/J 24/P23/Q4/a

4(a)	OR Correct representation or label showing M1 (ionic bond between) NH_4^+ and Cl^- AND 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 M2 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons	2
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25. MAR 24/P22/Q1/a

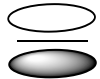
1(a)	labelled diagram shows: - regular arrangement of positive ions AND surrounded by / (sea of) delocalised electrons	1
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26. MAR 24/P22/Q2/c(i)

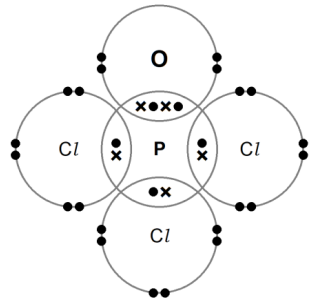
2(c)(i)	• trigonal bipyramidal • octahedral	2
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27. MAR 24/P22/Q3/d

3(d)(i)	correct bonding pairs	1
	lone pairs	1
3(d)(ii)	sp^2	1

3(d)(iii)	 M1 shows the pi (π) orbital	1
	M2 refers to overlap of (two) (2)p orbitals side-on / above and below the plane	1

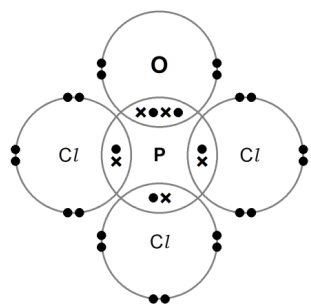
28. O/N 23/P21/Q1/d(iii)

1(d)(iii)	 Correct bonding e^-	1
	rest of electrons correct (i.e. to make 32 electrons in total)	1

29. O/N 23/P22/Q4/a(iii)

4(a)(iii)	dashed (or distinctly labelled) bond between O and H in different molecules of H-O-H and propanoic acid	1
	correct sequence of three partial charges over the H-bond $H^{\delta+} \quad O^{\delta-} \cdots H^{\delta+}$ OR $O^{\delta-} \quad H^{\delta+} \cdots O^{\delta-}$	1
	lone pair on O in line with H-bond	1

30. O/N 23/P23/Q1/d(iii)

1(d)(iii)	 Correct bonding e^-	1
	rest of electrons correct (i.e. to make 32 electrons in total)	1

31. M/J 23/P21/Q1/e

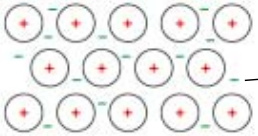
1(e)	M1 6 bonding pairs (of electrons and 0 lone pairs) OR coordination number = 6 OR $x = 6$ M2 bonding pairs repel (equally)	2
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32. M/J 23/P22/Q1/b

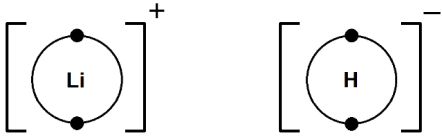
1(b)(i)	power of an atom to attract electrons to itself	1
1(b)(ii)	(across a period) • increase in nuclear charge • similar shielding • (so) increase in nuclear attraction for bonding / outer / valence electrons OR bonding / outer / valence electron(s) are more strongly attracted to nucleus Two correct for one mark, three correct for two marks	2

1(b)(iii)	hydrogen bond	1
1(b)(iv)	M1 link shown as a dashed line between the lone pair of electrons from N of one NH_3 to one H on other NH_3 M2 minimum 3 correct partial charges (on adjacent atoms) over two NH_3 molecules EITHER $\delta^- \text{N} - \delta^+ \text{H} \cdots \delta^- \text{N}$ OR $\delta^+ \text{H} \cdots \delta^- \text{N} - \delta^+ \text{H}$	2
1(b)(v)	M1 O is more electronegative than N M2 two H-bonds per water molecule : 1 per ammonia molecule.	2

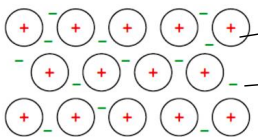
33. M/J 23/P23/Q1/a(ii)

1(a)(ii)	 delocalised electrons M1 diagram showing minimum of 4 particles - circles containing just a + do not have to be labelled - empty circles / circles with Cu must be labelled + ion / positive ion / cation / Cu^{n+} AND with the circles surrounded by electrons shown as e^- / – OR little circles labelled delocalised electrons	1
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34. MAR 23/P22/Q1/a,b,e(i)

1(a)(i)	power of an atom to attract electrons to itself	1										
1(a)(ii)	• O lower nuclear charge / lower proton number • O has (one) fewer shell than S / less shielding • greater attraction (for nucleus) in O	2										
1(b)(i)		2										
1(b)(ii)	non-linear	1										
1(e)(i)	<table><tr><td>compound</td><td>O=C=O</td><td>O=S=O</td><td>S=C=S</td><td>S=C=O</td></tr><tr><td>overall dipole moment</td><td></td><td>✓</td><td></td><td>✓</td></tr></table>	compound	O=C=O	O=S=O	S=C=S	S=C=O	overall dipole moment		✓		✓	2
compound	O=C=O	O=S=O	S=C=S	S=C=O								
overall dipole moment		✓		✓								

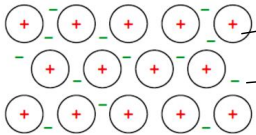
35. MAR 23/P22/Q2/a

2(a)(i)	 ion delocalised electrons M1 diagram showing minimum of 4 particles (in total in 2 rows) - circles containing X^{n+} do not have to be labelled - circles with '+' must be labelled as 'ion' OR empty circles / circles with X must be labelled + ion / positive ion / cation / X^{n+} AND with the circles surrounded by electrons shown as e^- / – OR little circles labelled electrons	1
	M2 label / legend showing delocalised electrons	1
2(a)(ii)	Mg has more delocalised e^- (than Na)	1

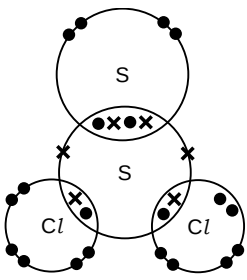
36. MAR 23/P22/Q3/d(iv)

3(d)(iv)	instantaneous dipole—induced dipole / id—id AND permanent dipole—permanent dipole / pd—pd	1
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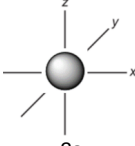
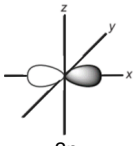
37. O/N 22/P21/Q2/a

2(a)	 M1 diagram showing minimum of 4 particles (in total in two rows) (circles) - circles containing Mg^{n+} do not have to be labelled - must be labelled as 'ion' OR empty circles / circles with Mg must be labelled + ion / positive ion / cation / Mg^{n+} AND - circles surrounded by electrons shown as e^- / – OR in an area around the circles labelled as 'electrons' OR little circles labelled electrons OR electrons drawn only on perimeter of structure	1
	M2 label / legend showing <u>delocalised</u> electrons	1

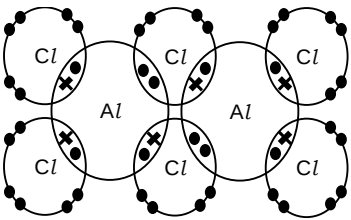
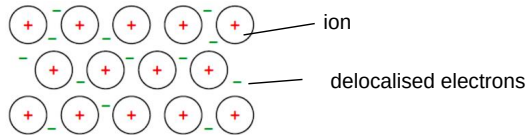
38. O/N 22/P21/Q3/d

3(d)(i)	M1 (structure =) simple / molecular, because it has a low melting / boiling point	1
3(d)(ii)	M1 ((-58.2) + (-40.6)) OR (-98.8) M2 [(-58.2 + 4(-40.6) = 8ΔH_f)] Correct calculation and correct stoichiometry ΔH_f = -27.6 (kJ mol⁻¹)	2
3(d)(iii)	equilibrium moves to left AND more moles / molecules of gas on LHS	1
3(d)(iv)	(molecules / isomers with) the same molecular formula	1
3(d)(v)	M1 90° < Cl—S—S ≤ 108° M2 sulfur has two lone pairs (of e⁻) (and two bonding pairs) AND repulsion from lone pairs (greater)	1
3(d)(vi)	 bonding electrons all other electrons correct	1

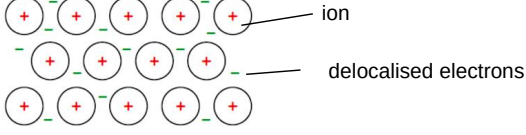
39. O/N 22/P22/Q1/a,c(iii)

1(a)	M1 ionic = oppositely charged ions / cations AND anions M2 covalent = shared pair(s) of electrons AND two nuclei	1
1(c)(iii)	 	2

40. O/N 22/P22/Q2/c(i),e(ii),f(i)

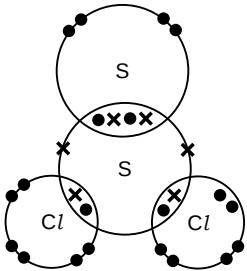
2(c)(i)		1
	'dot-and-cross' diagram to show M1 correct number and connectivity of Cl and Al atoms M2 2 correct dative bonds (••) between Cl and Al of $AlCl_3$ neighbour AND correct number of non-bonding electrons around each atom of Al_2Cl_6	1
2(e)(ii)	M1 sum of instantaneous dipole – dipole attractions and permanent dipole attractions in SCl_2 is greater than those in SBr_2 OR permanent dipole-permanent dipole (forces) in SCl_2 are stronger OR intermolecular forces are stronger in SCl_2 ora	1
	M2 SCl_2 (molecule / bond) is more polar than SBr_2 OR Cl is more electronegative than Br so SCl_2 has stronger (permanent) dipoles ora	1
2(f)(i)	 M1 diagram showing minimum of 4 particles (in total in 2 rows) - circles containing Bi^{n+} do not have to be labelled • must be labelled as 'ion' OR empty circles / circles with Bi must be labelled + ion / positive ion / cation / Bi^{n+} AND - with the circles surrounded by electrons shown as e^- / – OR little circles labelled electrons	1
	M2 label / legend showing <u>delocalised</u> electrons	1

41. O/N 22/P23/Q2/a

2(a)	 M1 diagram showing minimum of 4 particles (in total in two rows) (circles) - circles containing Mg^{n+} do not have to be labelled • must be labelled as 'ion' OR empty circles / circles with Mg must be labelled + ion / positive ion / cation / Mg^{n+} AND - circles surrounded by electrons shown as e^- / – - OR - in an area around the circles labelled as 'electrons' OR little circles labelled electrons - OR - electrons drawn only on perimeter of structure	1
	M2 label / legend showing <u>delocalised</u> electrons	1

42. O/N 22/P23/Q3/d

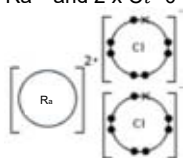
3(d)(i)	M1 (structure =) simple / molecular, because it has a low melting / boiling point	1
	M2 (bonding =) covalent, because it is hydrolysed	1

3(d)(ii)	Correct use of enthalpy values / correct direction / use of arrows (with or without an energy cycle) M1 $(-58.2) + (-40.6)$ OR (-98.8) M2 $[(-58.2) + 4(-40.6) = 8\Delta H_f]$ Correct calculation and correct stoichiometry $\Delta H_f = -27.6 \text{ (kJ mol}^{-1}\text{)}$	2
3(d)(iii)	equilibrium moves to left AND more moles / molecules of gas on LHS	1
3(d)(iv)	(molecules / isomers with) the same molecular formula	1
	but different structural formulae	1
3(d)(v)	M1 $90^\circ < \text{Cl-S-S} \leq 108^\circ$	1
	M2 sulfur has two lone pairs (of e^-) (and two bonding pairs) AND repulsion from lone pairs (greater)	1
3(d)(vi)	 bonding electrons	1
	all other electrons correct	1

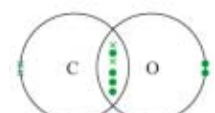
43. M/J 22/P21/Q2/a

2(a)	M1 one sigma / σ bond and two pi / π bonds M2 sp hybridisation (in each N atom) M3 sigma / σ forms from direct / head-on / end-on overlap of orbitals AND pi / π forms sideways / lateral overlap of (p) orbitals	3
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44. M/J 22/P22/Q2/b

2(b)	Ra^{2+} and $2 \times \text{Cl}^-$ 0 electrons surrounding 2Ra AND 8 electrons surrounding Cl 	1
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45. M/J 22/P23/Q3/a(i)

3(a)(i)		1
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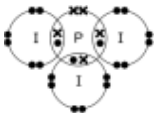
46. O/N 21/P21/Q2/c

2(c)(i)	Mixing / overlap / combination of one / an s and one / a p orbital	1
2(c)(ii)	Sketch a diagram to show HOW two sp hybrid orbitals can form a SIGMA bond M1  M2 	2

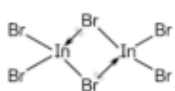
47. O/N 21/P21/Q4/d

4(d)	Forms hydrogen bonds with water	1
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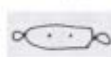
48. O/N 21/P22/Q1/a,f(i)

1(a)(i)	M1 (HI / I / iodine / hydrogen iodide has a) greater number of electrons M2 greater induced dipoles (between molecules)	2
1(a)(ii)	M1 bar at HF shows any boiling point above HI on graph M2 explanation of difference in boiling point of a sample of HF in terms of strength (sum of) hydrogen bonds (and induced dipoles in HF) are stronger than (sum of) induced dipoles (and permanent dipoles in HCl / HBr / HI)	2
1(f)(i)	 M1 correct bonding pairs M2 correct number of remaining outer electrons on each atom	2

49. O/N 21/P22/Q2/d(iii)

2(d)(iii)	 M1 correct connectivity of In ₂ Br ₆ M2 showing the two correct dative covalent bonds	2
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50. O/N 21/P23/Q2/c

2(c)(i)	Mixing / overlap / combination of one / an s and one / a p orbital	1
2(c)(ii)	Sketch a diagram to show HOW two sp hybrid orbitals can form a SIGMA bond M1  M2 	2

51. O/N 21/P23/Q4/d

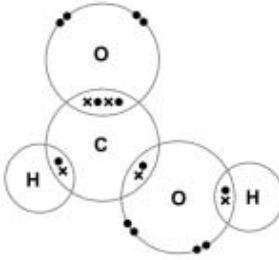
4(d)	Forms hydrogen bonds with water	1
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52. M/J 21/P21/Q2/a,b

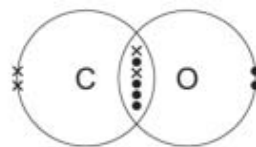
2(a)(i)	M1 both make triple (covalent) bond / 3 shared pairs of electrons			1
	M2 one bond in CO is coordinate / dative covalent / formed by donating a pair of electrons from O (to C)			1
2(a)(ii)		N ₂	CO	2
	number of electrons per molecule	14	14	
	type of van der Waals'	temporary / instantaneous dipole–induced dipole	permanent dipoles–(permanent) dipoles (and temporary / induced / instantaneous dipoles)	

2(b)	CO / it is a polar molecule / it has a (permanent) dipole (but N ₂ is non-polar)	1
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53. M/J 21/P21/Q4/d

4(d)	 M1 correct bonding electrons M2 correct number of non-bonding electrons around each oxygen	1
		1

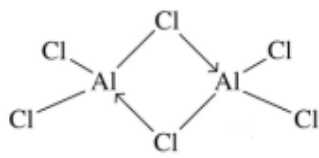
54. M/J 21/P22/Q2/a(i),b,c(iii,iv)

2(a)(i)	Li ⁺ AND S ²⁻	1				
2(b)(i)	(covalent) bond with both electrons are provided from the same / one species OR shared pair (of electrons) are provided from the same species / one atom <i>owtte</i>	1				
2(b)(ii)	3 bonding pairs between C and O, 4 •'s AND 2×'s 1 lone pair on C, ××, AND 1 lone pair on O, •• 	2				
2(c)(iii)	<table><tr><td>N₂(g)</td><td>CO(g)</td></tr><tr><td>instantaneous dipole–induced dipole ✓</td><td>instantaneous dipole–induced dipole (and) permanent dipole–permanent dipole ✓</td></tr></table>	N ₂ (g)	CO(g)	instantaneous dipole–induced dipole ✓	instantaneous dipole–induced dipole (and) permanent dipole–permanent dipole ✓	2
N ₂ (g)	CO(g)					
instantaneous dipole–induced dipole ✓	instantaneous dipole–induced dipole (and) permanent dipole–permanent dipole ✓					
2(c)(iv)	O is more electronegative than C	1				

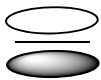
55. M/J 21/P23/Q2/a

2(a)	M1 H-bond between an O and an H in OH groups in water and methanol molecules	1
	M2 minimum three partial charges (in sequence) over one water and one methanol molecule, <i>i.e.</i> :	1
	M3 either $\delta^- \text{O} - \text{H}^{\delta+} \cdots \delta^- \text{O}$ or $\text{H}^{\delta+} \cdots \delta^- \text{O} - \text{H}^{\delta+}$ lone pair of electrons on O of H-bond, in line with H-bond	1

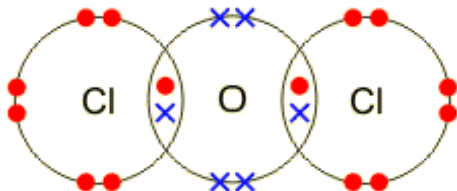
56. MAR 21/P22/Q2/e(i)

2(e)(i)	 M1: 2 × coordinate bonds in the right place M2: all other bonds	2
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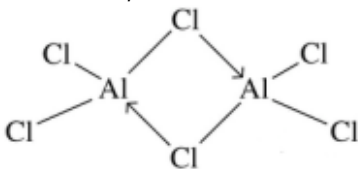
57. MAR 21/P22/Q3/b(ii)

3(b)(ii)	M1: overlap of two p orbitals side-on / above and below the plane M2:  pi (π) orbital	2
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58. O/N 20/P21/Q2/d(i)

2(d)(i)		1
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59. O/N 20/P21/Q3/a(i,ii,iv)

3(a)(i)	M1: correct representation of Al_2Cl_6, dot and cross or line diagram  M2: TWO correct co-ordinate bonds identified	2
3(a)(ii)	120	1
3(a)(iv)	(Lattice of) cations / positive ions surrounded by delocalised electrons'	1

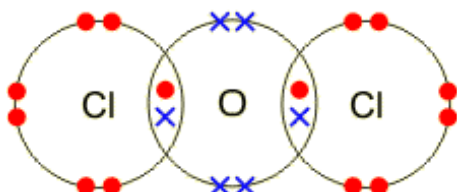
60. O/N 20/P21/Q4/a(iii)

4(a)(iii)	M1: HF has permanent dipole(-dipole forces) AND HI has ((only)) instantaneous dipole / induced dipole (forces) / permanent dipole(-dipole forces) M2: IMF's in HI are weaker (than IMF's in HF)	2
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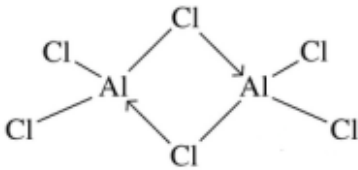
61. O/N 20/P22/Q3/d(i)

3(d)(i)	M1: (trigonal) pyramidal M2: 107	2
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62. O/N 20/P23/Q2/d(i)

2(d)(i)		1
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63. O/N 20/P23/Q3/a(i,ii,iv)

3(a)(i)	M1: correct representation of Al_2Cl_6, dot and cross or line diagram  M2: TWO correct co-ordinate bonds identified	2
3(a)(ii)	120	1
3(a)(iv)	(Lattice of) cations / positive ions surrounded by delocalised electrons	1

64. O/N 20/P23/Q4/a(iii)

4(a)(iii)	M1: HF has permanent dipole(-dipole forces) AND HI has ((only)) instantaneous dipole / induced dipole (forces) / permanent dipole(-dipole forces) M2: IMF's in HI are weaker (than IMF's in HF)	2
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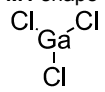
65. M/J 20/P21/Q1/b(ii)

1(b)(ii)	coordinate / dative (covalent)	1
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66. M/J 20/P21/Q4/c

4(c)	<table border="1"> <thead> <tr> <th></th><th>name of shape</th><th>bond angle / °</th></tr> </thead> <tbody> <tr> <td>CO₂</td><td>Linear</td><td>180</td></tr> <tr> <td>NH₃</td><td>Pyramid(al)</td><td>107</td></tr> <tr> <td>H₂O</td><td>non-linear / V / bent</td><td>104.5</td></tr> </tbody> </table> All 6 correct – 3 marks 4 or 5 correct – 2 marks 2 or 3 correct – 1 mark		name of shape	bond angle / °	CO ₂	Linear	180	NH ₃	Pyramid(al)	107	H ₂ O	non-linear / V / bent	104.5	3
	name of shape	bond angle / °												
CO ₂	Linear	180												
NH ₃	Pyramid(al)	107												
H ₂ O	non-linear / V / bent	104.5												

67. M/J 20/P22/Q1/d

1(d)	M1 shape  M2 bond angle 120(°)	2
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68. MAR 20/P22/Q1/d

1(d)	180(°)	1
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69. MAR 20/P22/Q2/d(i)

2(d)(i)	M1 increases (down the group) because of increasing VdW M2 because of increasing number of electrons	2
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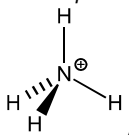
70. O/N 19/P21/Q2/c(i)

2(c)(i)	octahedral	1
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71. O/N 19/P21/Q4/b(ii)

4(b)(ii)	M1: A has H-bonding (between molecules) M2: B only has dipole–dipole / VdW forces (between molecules) M3: H-bonding is stronger / requires more energy to overcome	3
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72. O/N 19/P22/Q3/b(ii)

3(b)(ii)	3-D shape AND bond angle  AND $109\frac{1}{2}^{\circ}$	1
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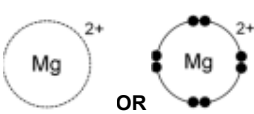
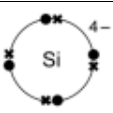
73. O/N 19/P23/Q2/c(i)

2(c)(i)	octahedral	1
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74. O/N 19/P23/Q4/b(ii)

4(b)(ii)	M1: A has H-bonding (between molecules) M2: B only has dipole–dipole / VdW forces (between molecules) M3: H-bonding is stronger / requires more energy to overcome	3
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75. M/J 19/P21/Q2/a,d

2(a)	M1 magnesium +2 charge on two Mg AND both with 0 or 8 electrons  OR 	1
	M2 silicide -4 charge on one Si and 8 electrons	1
2(d)(i)	C δ^- –H δ^+	1
	Si δ^+ –H δ^-	1
2(d)(ii)	M1 tetrahedral (molecule)	1
	M2 (so individual bond) dipoles / partial charges cancel	1

76. M/J 19/P22/Q2/c(iii,iv)

2(c)(iii)	co-ordinate / dative (covalent)	1
2(c)(iv)	(triangular / trigonal) pyramid(al)	1
	107 ⁽⁰⁾	1

77. M/J 19/P22/Q3/a(iv)

3(a)(iv)	M1 statement correctly comparing the difference in electronegativity between Si and Cl AND Na and Cl OR Na is less electronegative than Si <i>ORA</i>	1
	M2 NaCl transfer of electrons	1
	M3 SiCl ₄ shared (pair of) electrons	1

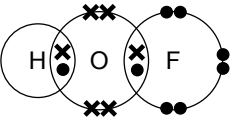
78. M/J 19/P23/Q2/b

2(b)(i)	2	1
2(b)(ii)	iodine (atom / s) donates a pair of electrons (to the Al-I covalent bond / s).	1

79. MAR 19/P22/Q1/d(iii)

1(d)(iii)	M1 $+82 (= E_{O=O} - 2E_{N-O}) = (+)496 - 2 \times E_{N-O}$ M2 $E_{N-O} = \frac{1}{2} \times (496 - 82) = \frac{1}{2} \times 414 = 207 \text{ (kJ mol}^{-1}\text{)}$	2
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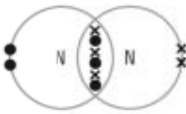
80. MAR 19/P22/Q2/a(ii),c(i,iv)

2(a)(ii)	M1 identification of the IMF between F_2 molecules and between HCl molecules HCl has (permanent) dipoles and / or induced dipoles F_2 has induced dipoles M2 comparison of strength of IMF's in F_2 and HCl Intermolecular forces in HCl are stronger than F_2	2
2(c)(i)	 M1 bonding pairs correct M2 rest of molecule, incl. lone pairs.	2
2(c)(iv)	$\delta+ \quad \delta- \quad \delta+ \quad \delta-$ $H-F \vdots \vdots \vdots \vdots H-F$ M1 H-bond labelled / shown as distinct from $H-F$ bond M2 correct sequence of three correct dipoles M3 lone pair on F in line with H-bond	3

81. O/N 18/P21/Q1/b

1(b)	M1 attraction/hold M2 positive ions / cations AND delocalised electrons (may be seen in a labelled diagram)	2
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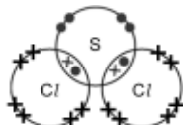
82. O/N 18/P21/Q2/a(ii)

2(a)(ii)	 6 e^- between atoms AND two electrons on each N atom	1
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83. O/N 18/P22/Q1/b

1(b)	metallic	1
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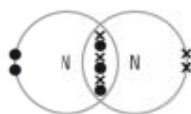
84. O/N 18/P22/Q2/b(i)

2(b)(i)		1
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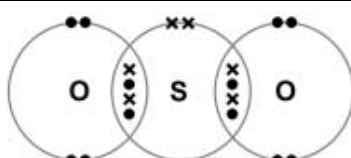
85. O/N 18/P23/Q1/b

1(b)	M1 attraction/hold M2 positive ions / cations AND delocalised electrons (may be seen in a labelled diagram)	2
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86. O/N 18/P23/Q2/a(ii)

2(a)(ii)	 6 e⁻ between atoms AND two electrons on each N atom	1
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87. M/J 18/P21/Q1/f(i)

1(f)(i)		1
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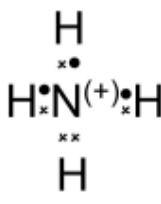
88. M/J 18/P21/Q2/a

2(a)	Different (hydrocarbon) molecules have different numbers of electrons	1
	so different strengths / numbers / amount of VdW / IMFs / id-id	1

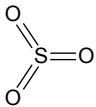
89. M/J 18/P22/Q1/b(iv)

1(b)(iv)	<i>any two from:</i> - increasing (cation) size / (cat)ionic radius increases / increasing size of atoms - overall charge on (cation) is constant / (+)2 / decreasing charge density of (+2 charged cation) - number of delocalised / outer / valence electrons (per atom) remains the same / 2 - increased shielding	2
	decreasing (strength of) attraction between (cat)ion / nucleus / nuclear charge AND delocalised electron(s)	1

90. M/J 18/P22/Q2/b

2(b)(i)		
	4 shared pairs only (any symbols) (in NH ₄)	1
	3 × dot-and-cross bonds AND 1 × 2 crosses (in NH ₄)	1
2(b)(ii)	tetrahedral	1
	109–109.5° (inclusive)	1

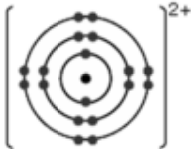
91. M/J 18/P23/Q1/a

1(a)	 	4
	trigonal planar non-linear/bent/V-shaped/angular	

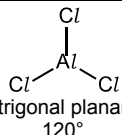
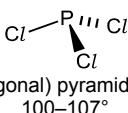
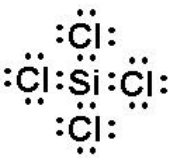
92. MAR 18/P22/Q2/d(iv)

2(d)(iv)	tetrahedral	1
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93. MAR 18/P22/Q3/b(i)

3(b)(i)		1
	dot-and-cross diagram AND 2+	

94. O/N 17/P22/Q1/a,b(ii,iii)

1(a)	 trigonal planar 120°  (trigonal) pyramid(al) 100–107° 3 marking points for each box: diagram, name and shape. for each box: all three correct = 2 marks two correct = 1 mark	4
1(b)(ii)	SiCl ₄ has more electrons ORA	1
	stronger Van der Waals' / id-id forces / London / dispersion forces / IMFs	1
1(b)(iii)		1

95. O/N 17/P22/Q3/b(ii)

3(b)(ii)	increased metallic / (cat)ionic radius / size OR decreasing (cat)ion charge-density	1
	decreased attraction (of ions) for delocalised / outer electrons	1

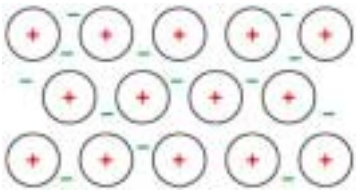
96. M/J 17/P21/Q2/b

2(b)(i)	hydrogen bonding	1
2(b)(ii)	H-bond between O and H of different molecules	1
	minimum three partial charges (in a row) over two H₂O molecules, i.e.: either $\delta^- \text{O} - \text{H}^{\delta+} \text{ --- } \delta^- \text{O}$ or $\text{H}^{\delta+} \text{ --- } \delta^- \text{O} - \text{H}^{\delta+}$	1
	lone pair of electrons on O of H-bond, in line with H-bond	1

97. M/J 17/P22/Q4/a(ii)

4(a)(ii)	(A / straight chain) has strong(er) (temporary dipole-) induced dipole (attractions) <i>ora</i>	1
	(because A / straight chain has) bigger (surface) area / more (points of) contact (in unbranched isomer) <i>ora</i> OR (so) more energy required to break the intermolecular forces <i>ora</i>	1

98. MAR 17/P22/Q1/c(i)

1(c)(i)	 diagram showing regular arrangement of (positive) ions surrounded by / sea of (delocalised) electrons	2
		1
		1

99. MAR 17/P22/Q2/a(i,ii)

2(a)(i)	bond in which the centres of positive and negative charges do not coincide OR electron distribution is asymmetric / unequal OR two (bonded) atoms are partially charged	1
2(a)(ii)	HF has the strongest (permanent) dipole–dipole / van der Waals' (forces) / HF has hydrogen bonding	1
	requires more energy to overcome (than weaker (permanent) dipole–dipole / van der Waals' forces between other hydrogen halides)	1

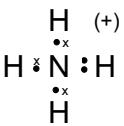
100. M/J 16/P22/Q2/b

(b) (i)	stronger / more / greater id–id / London / dispersion forces	[1]	[2]
	due to increasing numbers of electrons	[1]	
(ii)	(intermolecular forces in HF are) hydrogen bonds (which are) stronger (than vdW) / more energy needed to separate molecules	[1] [1]	[2]
	OR HF much more polar / F much more electronegative Intermolecular forces in HF stronger (than in HCl, HBr, HI)	[1] [1]	

101. M/J 16/P23/Q1/b(i)

(b) (i)	covalent small(er) difference in electronegativity between Fe and Cl (than between Al and Cl)	[1] [1]	[2]
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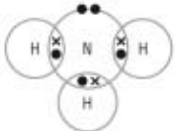
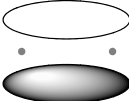
102. M/J 16/P23/Q2/d

(d) (i)		[1+1]	[2]
(ii)	shape = tetrahedral angle = 109°–109.5°	[1] [1]	[2]

103. MAR 16/P22/Q1/a(iv)

(iv)	magnesium has <u>two</u> free / delocalised / outer / valence electrons per atom OR <u>more</u> free / delocalised / <u>outer</u> electrons than sodium	[1]	[1]
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104. MAR 16/P22/Q3/a

3 (a) (i)	 three bonding pairs lone pair AND octet shape = (trigonal) pyramidal	[1] [1] [1]	[3]
(ii)	sigma(σ) bond  OR  pi(π) bond 	[1] [1]	[2]

105. MAR 16/P22/Q4/d

(d)	M1 = H has more / greater / stronger van der Waals' / intermolecular forces than G / ora M2 = (because) H has more electrons (than G) M3 = J has hydrogen bonding (between molecules) M4 = strong(er) / great(er) forces require AND high / more energy to overcome	[1] [1] [1] [1]	[4]
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106. O/N 15/P22/Q1/b,c

(b) (i)	ability / tendency / power of an atom / nucleus to attract / pull electron(s) in a covalent bond / shared pair of electrons / bonding pair of electrons	[1] [1]	[2]
(ii)	Covalent overlap of orbitals OR shared <u>pair(s)</u> (of electrons) OR metallic positive ions / cations surrounded by delocalised electrons	[1] [1] [1] [1]	[2]
(iii)	Ionic / electrovalent (electrostatic) Attraction between oppositely charged / +ve and -ve <u>ions</u>	[1] [1]	[2]
(c) (i)	similar strength / amount / number of intermolecular forces / induced dipole / van der Waals' / VdW / London forces / LDF / dispersion forces therefore similar energy needed	[1] [1]	[2]
(ii)	M1 HCl polar / has a dipole AND F₂ non-polar / has no dipole OR (permanent) dipole (-dipole) attractions / forces between HCl (molecules) AND induced dipole (-induced dipole) attractions / forces / LDFs between F₂ (molecules) M2 more energy needed for HCl than F₂ OR pd-pd forces stronger than id-id forces OR IMFs / VdWs in HCl stronger than in F₂	[1] [1]	[2]
(iii)	Hydrogen bonding (between methanol molecules) Stronger than IMFs / van der Waals' in other three / is the strongest intermolecular force	[1] [1]	[2]

107. M/J 15/P23/Q1/e

1 (e) (i)	Van der Waal's/London/dispersion Uneven electron distribution/temporary dipole Induced dipole-dipole attraction	[1] [1] [1]	[3]
(ii)	more electrons more polarisable/greater attraction/stronger IMFs	[1] [1]	[2]